

# From Demand Forecasts To Replenishment Decisions

## **Demand Forecasting & Inventory Analytics**

UE Digital Analytics im Handel, JKU Linz · Session T2, 03.10.2026 · Sina Mirshahi

# About Me

- I am **Sina Mirshahi**, a researcher at Logistikum, University of Applied Sciences Upper Austria, in Steyr
- I work on applied AI and predictive analytics with industrial and logistics partners: forecasting, inventory, computer vision, and AI assistants that answer from company documents
- I am a Ph.D. candidate in Engineering Informatics at Tomas Bata University, where I taught neural networks and data science
- Recent papers: Forecasting Without Fine-Tuning, ready-trained forecasting models against traditional methods (Procedia Computer Science 277, 2026); Two-Step Framework for Intermittent Demand Forecasting (Procedia Computer Science 278, 2026)

Questions are always welcome, here or by email: [Sina.Mirshahi@fh-steyr.at](mailto:Sina.Mirshahi@fh-steyr.at)

# Where This Afternoon Fits

| Session                     | Lecturer    | The question                                                                           | What you leave with                                                                        |
|-----------------------------|-------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| T1, 03.10. morning          | Patrick     | What are real life use cases for retail analytics, and what are the data requirements? | Understanding of the importance of analytics in retail and the relevance of the clean data |
| <b>T2, 03.10. afternoon</b> | <b>Sina</b> | <b>How do we forecast demand and derive the order quantity?</b>                        | <b>A forecast choice and a service target, each with one number and one limitation</b>     |
| T3, 04.12.                  | Taha        | Which orders may come back, and what should we do?                                     | Return-risk prediction and the cost of prediction errors                                   |

- This afternoon has four chapters: understand the demand, forecast it, evaluate the forecast, turn it into stock. Two hands-on blocks in Colab, one break

→ **Today you turn a retail forecast into an order and a service target you can defend.**

# What You Will Take Home

Rather than coding from scratch, we look at how a decision becomes code: a rule the computer runs and you can read and question.

- **Recognise** what kind of demand a product has, and know what error to expect and which methods to try before anyone builds a model
- **Know the main methods**, from repeating last year to smoothing, methods for products that sell rarely, and one model for the whole assortment, and what each can and cannot see
- **See sales as model inputs**: how weekly sales become the numbers a model learns from, and how a model's settings are tuned
- **Read an accuracy number** and know what it hides: why the most common one fails on zeros, why a useless forecast can score well, why the direction of the error matters as much as its size
- **Ask how a forecast was tested**: how many test dates, how far ahead, with which metric, and whether the future was allowed in
- **Understand a model's reasons** for one forecast, as factors you can check against your own experience
- **Turn a forecast into stock**: how large a cushion to hold, how much to order, and what a higher service target buys

→ **Reading forecasts and acting on them, with the mechanics understood, not only the vocabulary.**

# Google Colab And Gemini

A notebook in your browser: text, code and results on one page, run on Google's computers. Free with a Google account.

- **A notebook** is a page of cells: text cells explain, code cells run and show their result right below
  - **Colab** runs the code on a Google computer in the cloud: nothing to install, and your copy is saved in your Google Drive
  - **Run** one cell with the play button next to it, or Shift + Enter; **Runtime, Run all** runs the whole notebook from the top
  - **Gemini**, Google's AI assistant, is built in if your account offers it: ask in plain English to explain an output or an error, or to write code from a description
- Check Gemini against the numbers: it can sound sure and still be wrong. Give it only the course data, never company or personal data

| Other notebook tools         | Where it runs                        | Good for                                         |
|------------------------------|--------------------------------------|--------------------------------------------------|
| Jupyter Notebook, JupyterLab | your own computer; free, open source | the original; Colab uses its file format, .ipynb |
| Kaggle Notebooks             | the browser; free, also from Google  | public datasets and competitions                 |
| VS Code with Jupyter         | your own computer; free              | notebooks next to regular programs               |
| Databricks, Microsoft Fabric | company cloud platforms              | notebooks on a company's own data                |

*The same .ipynb notebook file opens in all of them.*

## Try it now, before the course notebook

Open [colab.research.google.com](https://colab.research.google.com) and sign in, choose **New notebook**, type 2 + 2 in the code cell and press **Shift + Enter**. Then open **Gemini** and ask: "What does a code cell do?"

- **A notebook keeps the words, the code and the results on one page; Gemini helps you read it, and you check Gemini.**

# Loading Data Into Colab

Three ways to get a file into Colab. Each ends in one line of code that reads the file into a table.

| Route                        | What you do                                                                    | The code                                                                                                                        | Keep in mind                                                                    |
|------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>1. Upload a file</b>      | Files panel (the folder icon on the left), upload, pick the file               | <code>pd.read_csv("sample_weekly_sales.csv")</code>                                                                             | The file is deleted when the session ends                                       |
| <b>2. From a direct link</b> | Paste the address of the file itself; on GitHub, the Raw button                | <code>pd.read_csv("https://raw.githubusercontent.com/.../sample_weekly_sales.csv")</code>                                       | A Drive share link is not a direct link. Our notebooks load their data this way |
| <b>3. From Google Drive</b>  | Put the file in a Drive folder, then connect Drive (Colab asks for permission) | <code>drive.mount("/content/drive")</code><br><code>pd.read_csv("/content/drive/MyDrive/JKU_T2/sample_weekly_sales.csv")</code> | The file stays in your Drive between sessions                                   |

- **Then explore:** `df.head()` shows the first rows, `df.describe()` the numbers at a glance. Or ask Gemini to summarise and plot the table, and check what it did

→ Upload only data you are allowed to share: files in Colab and in your Drive sit on Google's servers

Try all three with the sample file (three products, weekly store sales): [notebook 0 in Colab](#).

→ **A file on your computer: upload it. A file on the web: use its link. A file you want to keep: put it in Drive.**

# Open Your First Notebook Now

The notebooks and the data live at [github.com/FH-Prevail/jku\\_course](https://github.com/FH-Prevail/jku_course). You need a Google account. No account, or it does not work? Work with a partner.

- **Open:** go to the repository (link on the board, or scan the code) and click the **Open in Colab** button next to notebook 1 (**1 · Demand Patterns and Forecast Comparison**) in the table on that page
  - Nothing happens? Open the notebooks folder, click **1\_see\_and\_forecast.ipynb** and use the button at its top, or in Colab: **File, Open notebook, GitHub**, type FH-Prevail/jku\_course
- **Make it yours:** click **Copy to Drive** (or File, Save a copy in Drive). Your copy keeps whatever you change
- **Run it:** Runtime, **Run all**. The first cell, **Start here**, fetches the course data; after a few seconds it prints **Ready** right under itself
- **Read:** the tables and pictures are the lesson. Use the dropdowns and sliders; nothing else needs editing
- **Ask:** Gemini explains any output ("explain this table in plain English"); check it against the numbers
- Accidentally changed a cell? Edit, Undo, then run it again. Restarting does not undo edits; a fresh copy from the repository does



*The repository: the course folder on GitHub*

## What you should see under the first cell (Start here) after Run all

Ready. The course data is loaded.  
The forecasts were computed in advance, using only what was known at each forecast date.

→ **Everyone should see the word Ready before we go on.**

# Understand The Demand

Channels, seasons, promotions,  
empty shelves, and four kinds of  
demand

# 01



# Meet Our Retailer

Artificial data, built like a real retailer's: one row per product, channel and week.

| What we have                                     | What it means                                                    |
|--------------------------------------------------|------------------------------------------------------------------|
| <b>40 products in 2 channels</b>                 | 80 series: each product in store and online                      |
| <b>156 weeks, 518,571 units</b>                  | 2023-01-02 to 2025-12-22, weeks starting Monday                  |
| <b>Plush Bear, Puzzle 1000, Vacuum Filter</b>    | Our three recurring examples: regular, variable and sparse sales |
| <b>1,037 promotion weeks with their discount</b> | Planned in advance: information we have before we order          |
| <b>4 flagged stock-out weeks</b>                 | Sales were zero because the shelf was empty; demand was not zero |

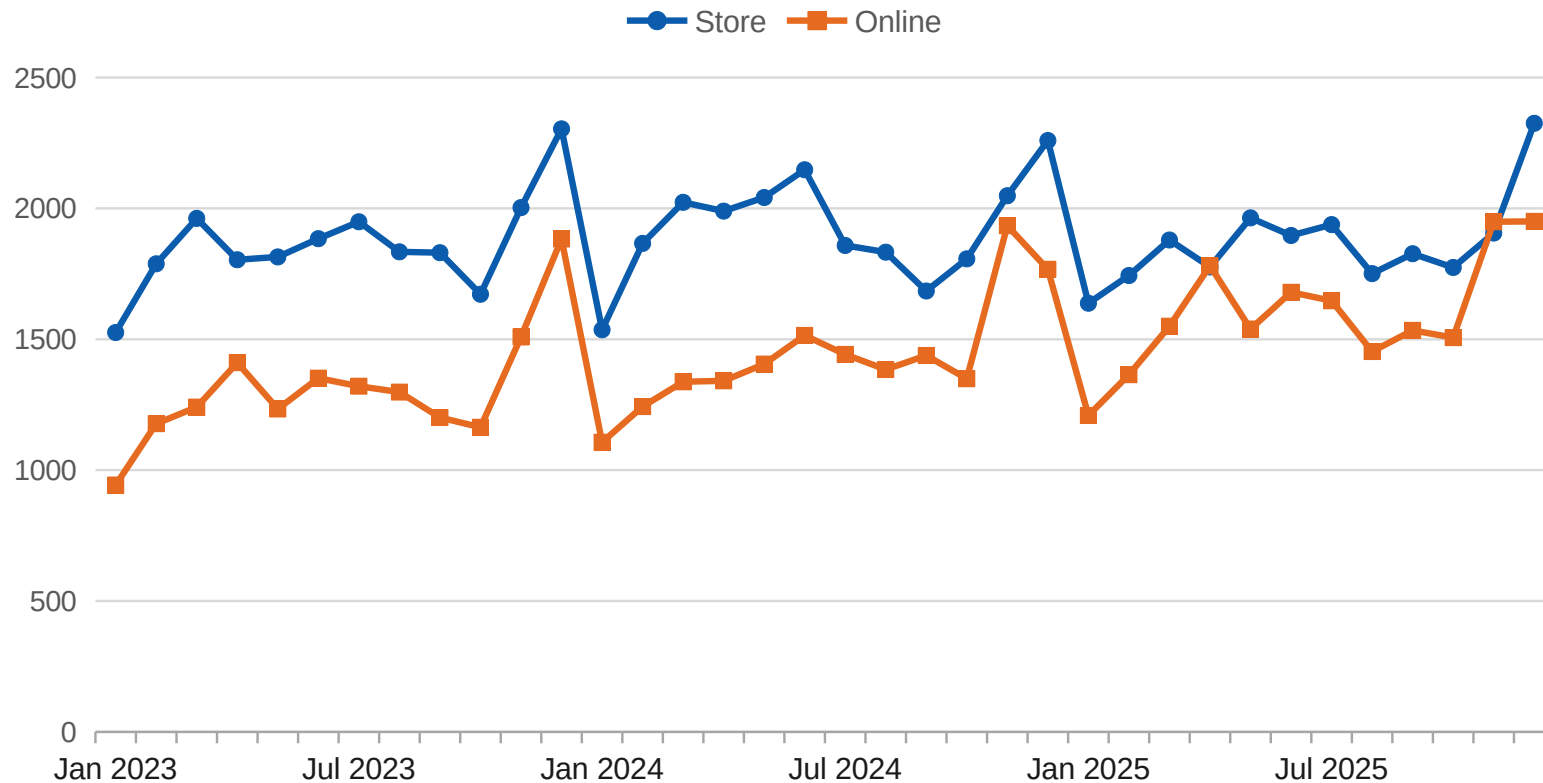
*Three rows of data/classroom/demand.csv. The first two: one product, one week, two channels, so two series. The third: a planned promotion, 15 % off.*

| week_start | product_name | channel | units | price | discount_pct | promo_flag | stockout_flag |
|------------|--------------|---------|-------|-------|--------------|------------|---------------|
| 2023-01-02 | Plush Bear   | store   | 77    | 25.00 | 0            | 0          | 0             |
| 2023-01-02 | Plush Bear   | online  | 57    | 25.00 | 0            | 0          | 0             |
| 2023-04-24 | Plush Bear   | store   | 177   | 21.25 | 15           | 1          | 0             |

→ A product in a channel is one series of weekly sales. We forecast series.

# Sales By Channel

All 40 products together, average units sold per week in each month, 2023 to 2025.



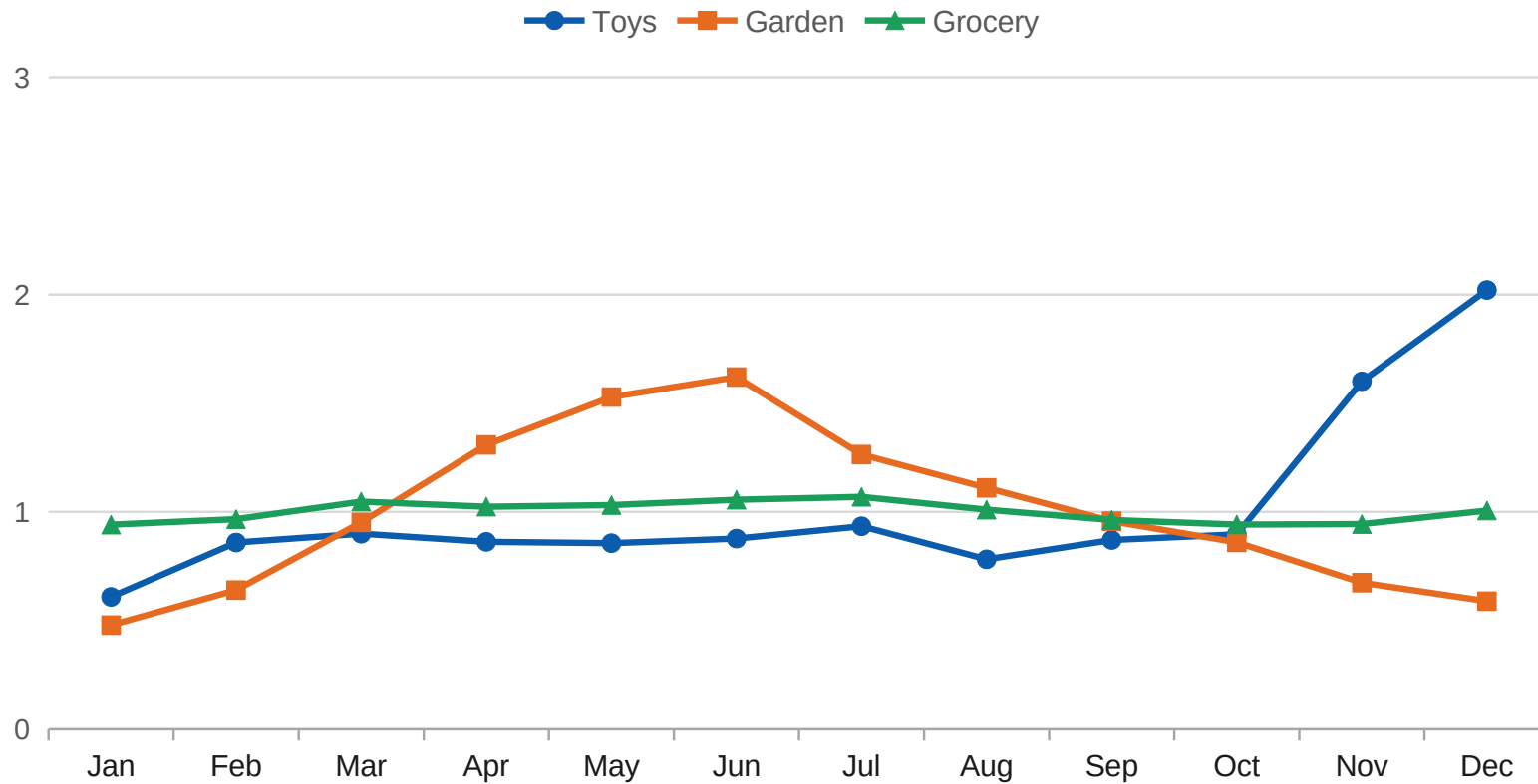
| Year | Store | Online |
|------|-------|--------|
| 2023 | 1,855 | 1,300  |
| 2024 | 1,919 | 1,437  |
| 2025 | 1,868 | 1,596  |

Average units per week in each year. Online grows every year; the store moves up and down. A single product can look different from its channel total.

→ Do not assume that both channels follow the same trend.

# The Busy Season By Category

Seasonal index: a month's average weekly sales divided by the category's average weekly sales over all three years, 2023 to 2025. 1.00 is the category's average month.

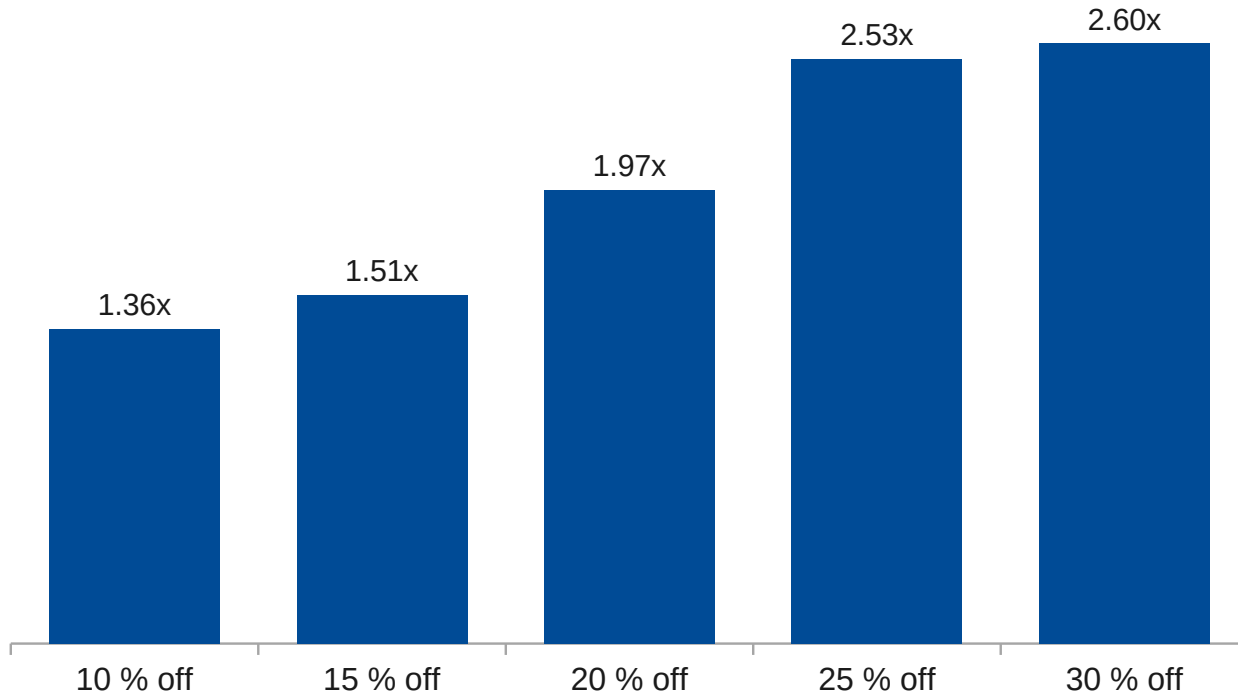


- **Toys** sell 2.0x their average month in December (Electronics, not shown: 1.5x)
- **Garden** peaks in June at 1.6x and drops to 0.5x in January
- **Grocery** stays between 0.94 and 1.07: almost no season
- A method that cannot see the calendar will be surprised every December

→ When the pattern repeats every year, the forecast should expect it.

# Promotion Effects By Discount

Sales in a promotion week compared with a normal week of the same product, by discount. 1.00x is a normal week.



*Normal week: no promotion this week or the week before.*

→ **Planned promotions belong in the information the forecast is given.**

- A deeper discount sells more: from **1.36x** at 10 percent to **2.60x** at 30 percent
- Online reacts more than the store: **2.10x** against **1.91x**
- The week after a promotion sells **9 percent below normal**: a sign that some customers bought earlier, during the promotion
- Promotions are **planned**, so a forecast can know about them in advance. How much of the extra sales is new demand, and how much only moved forward, needs a separate test

# Zero Sales And Empty Shelves

| What we observe               | What we know                                                                         | What to do                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| No sale, stock on the shelf   | Demand was zero                                                                      | Keep the week as it is                                                              |
| No sale, shelf empty          | Demand is unknown                                                                    | Flag the week; estimate it from the weeks around it, or leave it out of the scoring |
| No sale recorded, broken data | Nobody knows what sold: a store closed, a data feed failed, a product got a new code | Fix the data, or mark the week as missing; never keep it as a zero                  |
| Future stock availability     | Nobody knows it in advance                                                           | Never use a stock-out flag as an input for future weeks                             |

- In our data **4 weeks** are flagged. When the models learn from the history (training), each one is replaced by the average of the four weeks before it; when we measure forecast errors (scoring), they are left out
- Real sales data rarely has this flag: the till records what was sold, not whether the shelf was empty. Check the stock records first
- A quick test is the other channel: the 4K Streaming Stick sold nothing online for two weeks in February 2025, while the store sold 27 and 5

→ **Zero sales can mean no demand, no stock or no data. Tell them apart before you forecast.**

# Two Numbers Per Series

ADI (average demand interval): how often a product sells.  $CV^2$  (squared coefficient of variation): how much the size of a sale varies. A teaching example with 12 made-up weeks.

## ADI: how often it sells

weeks  $\div$  weeks with a sale

Sells every week:  $12 \div 12 = \text{ADI } 1.0$



Sells in 4 of 12 weeks (dots: no sale):  $12 \div 4 = \text{ADI } 3.0$



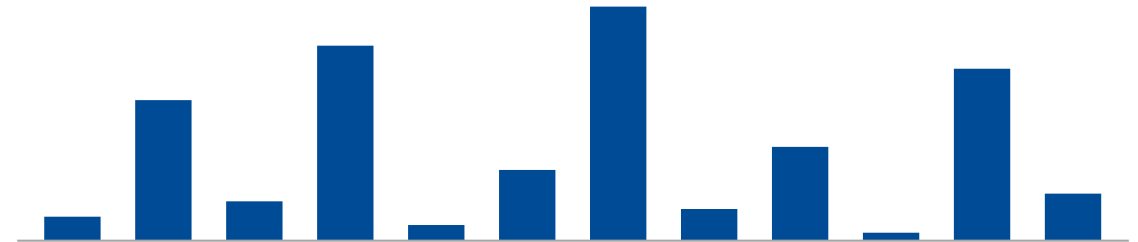
## $CV^2$ : how much the size of a sale varies

(spread of the sizes  $\div$  their average)<sup>2</sup>, zero weeks left out

Similar sizes, about 10 each week:  $CV^2$  0.01



Very different sizes, 1 to 30:  $CV^2$  0.69



→ **ADI counts the weeks per sale;  $CV^2$  compares the sizes of the sales, and only the weeks with a sale.**

# Four Demand Patterns

Two limits sort every series into one of four patterns: ADI 1.32 and  $CV^2$  0.49.

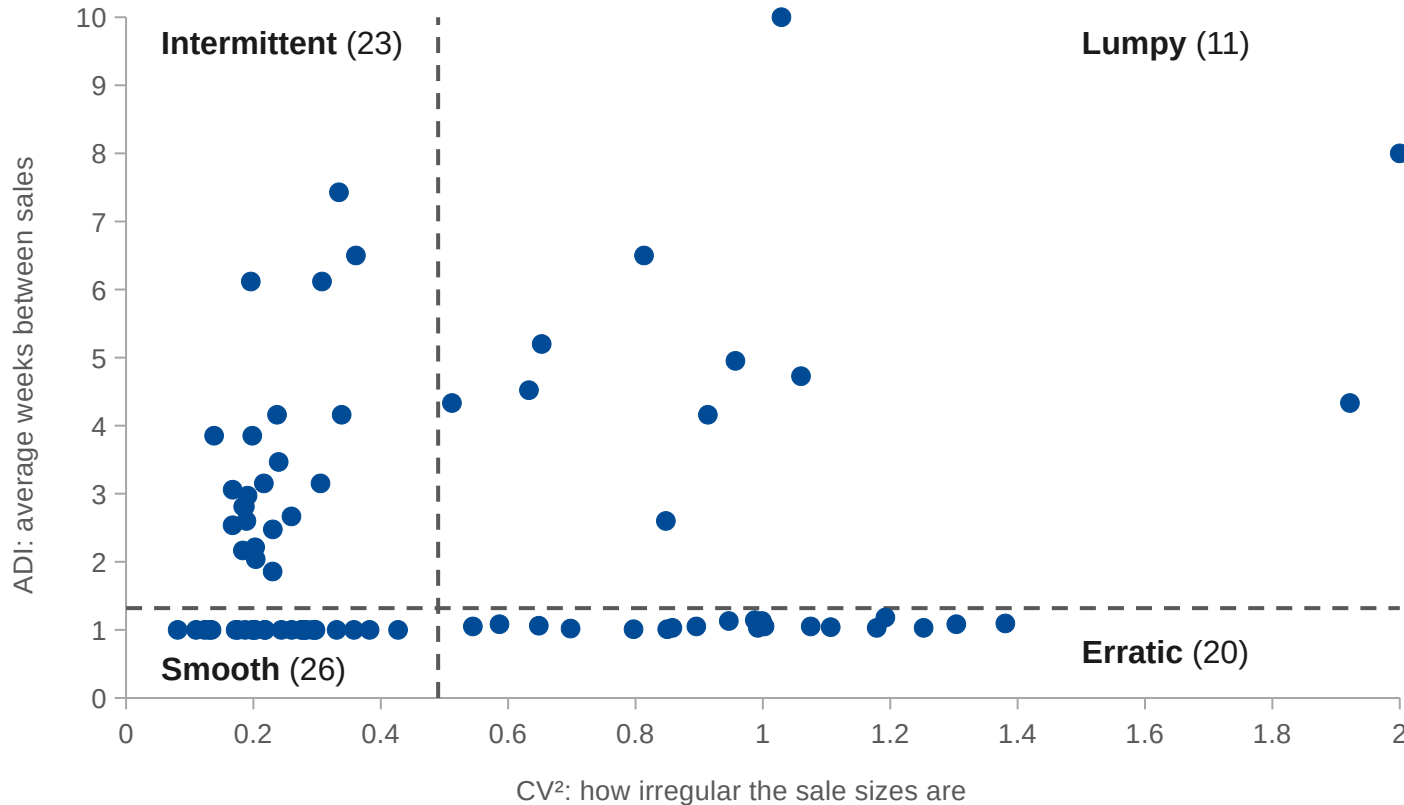
|                                                     | $CV^2$ below 0.49: regular sizes            | $CV^2$ at least 0.49: irregular sizes |
|-----------------------------------------------------|---------------------------------------------|---------------------------------------|
| <b>ADI at least 1.32: many weeks without a sale</b> | INTERMITTENT: 23 series, 0.4 % of the units | LUMPY: 11 series, 0.3 % of the units  |
| <b>ADI below 1.32: a sale most weeks</b>            | SMOOTH: 26 series, 89 % of the units        | ERRATIC: 20 series, 11 % of the units |

- The limits 1.32 and 0.49 are a convention from the literature (Syntetos and Boylan), not a law. You still test which method works
- **In our own research the limits are adaptive:** they are refined on a company's own data, with forecast accuracy as the guide, so more items count as smooth while the errors stay acceptable ([Mirshahi et al. 2025, ACDSA](#))
- 34 of 80 series are slow movers: intermittent or lumpy. In 2025 they sold under one percent of the units. They have methods of their own (Croston, SBA, TSB; reference slides at the end). The classes use only 2023 and 2024

→ **The pattern tells you what error to expect and which method to try first; it does not choose the method for you.**

# Four Patterns, One Picture

Every series placed by its two numbers, measured on 2023 and 2024. The dashed lines are the conventional limits:  $CV^2$  0.49 and ADI 1.32.



- **Each dot is one product in one channel** (80 dots; the two values outside the picture are drawn on its edge)
- **Further right**: the amounts sold vary more from one sale to the next
- **Further up**: more weeks pass between sales. ADI 1 means a sale every week; ADI 4 means a sale every four weeks on average
- The fast movers crowd the bottom edge; the slow movers sit above the dashed line, most between ADI 2.5 and 5

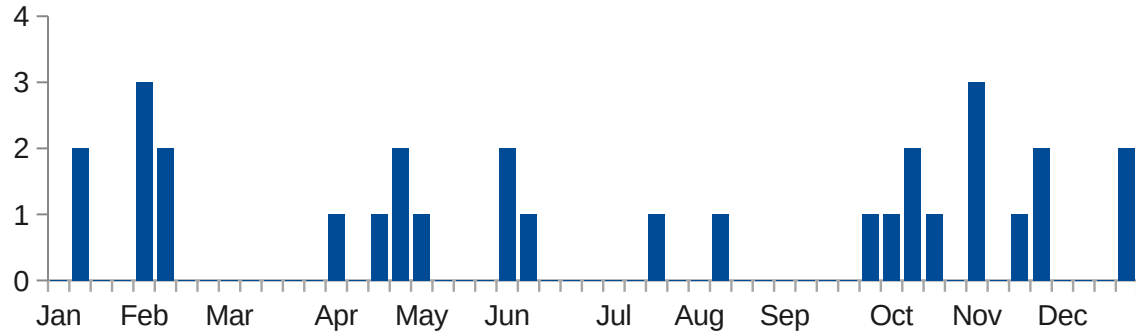
→ The higher a dot sits, the less a weekly forecast can tell you.



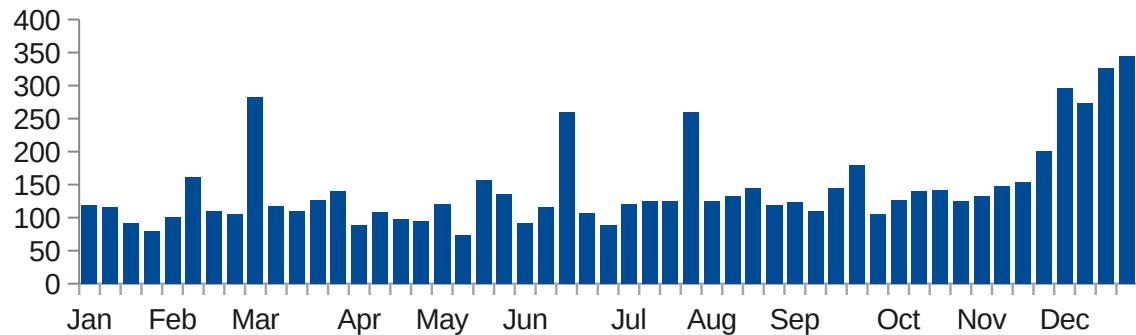
# Pattern Examples

Weekly units in 2025 for one product of each kind, placed in the same corners as on the picture.

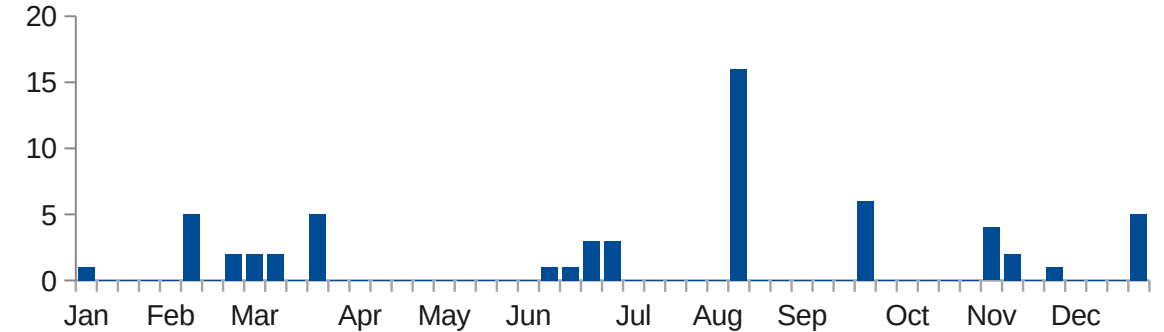
**Intermittent:** Vacuum Filter, store. ADI 2.54,  $CV^2$  0.17 (2023 and 2024); in 2025 63 % zero weeks, 0.6 units a week



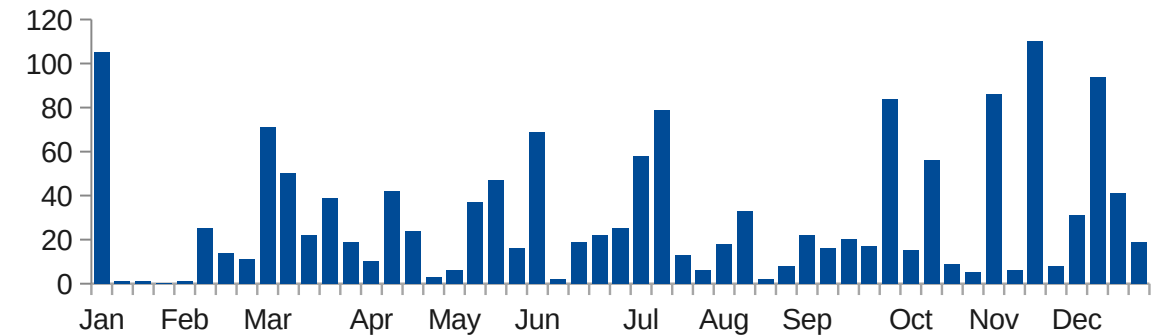
**Smooth:** Plush Bear, store. ADI 1.00,  $CV^2$  0.30 (2023 and 2024); in 2025 0 % zero weeks, 145.4 units a week



**Lumpy:** Trimmer Blade, store. ADI 2.60,  $CV^2$  0.85 (2023 and 2024); in 2025 69 % zero weeks, 1.2 units a week



**Erratic:** Puzzle 1000, store. ADI 1.03,  $CV^2$  1.18 (2023 and 2024); in 2025 2 % zero weeks, 30.1 units a week



# Forecast

From simple rules to a local model  
and a global model, compared on  
the same four weeks

02

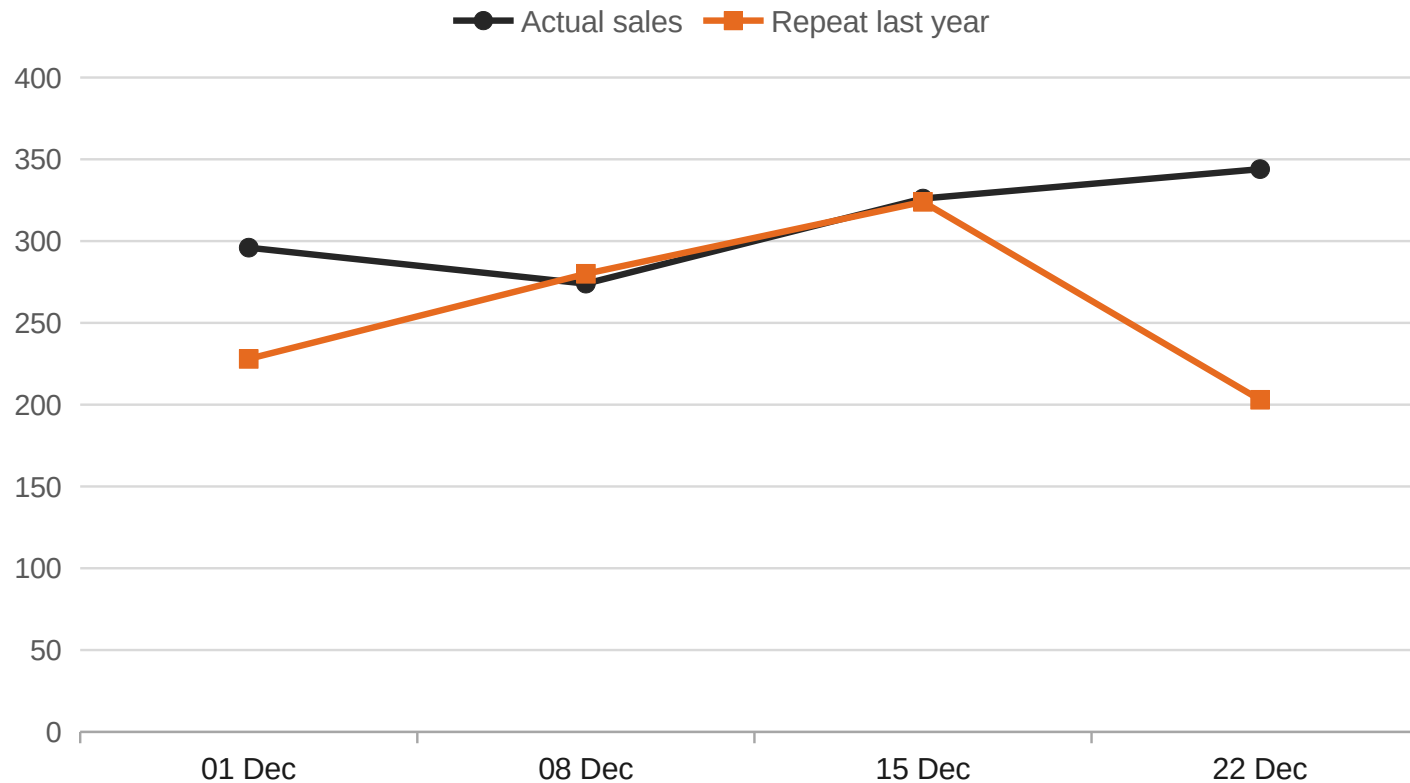
# Four Approaches Worth Comparing

| Approach                                                                                   | What it uses                                                                                   | Main limitation                                                           |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Repeat last year<br>(seasonal naive)                                                       | The same week 52 weeks earlier                                                                 | Repeats last year's unusual weeks too, and cannot see a changed promotion |
| Simple exponential smoothing<br>(SES)                                                      | A weighted average that trusts the newest weeks most; the forecast is a flat level             | This version does not anticipate a season                                 |
| Prophet<br>(a local model from Meta)                                                       | The product's own history as a trend, a yearly season and the planned discount, added together | Learns from one product only; overreacts on irregular sales               |
| LightGBM in global mode<br>(many small decision trees; one model trained on all 80 series) | Shared history of all products, plus calendar, product identity, planned price and promotion   | More preparation, harder to explain, and a win is not guaranteed          |

- The first three are **local**: each series is forecast from its own past. The fourth runs in **global mode**: one model learns from all series at once. LightGBM can also be trained per series, in local mode
  - The next slides show them on the Plush Bear's December weeks. In chapter 3 a replay of 2025, the backtest, scores all of them, plus Holt-Winters, the average of the last 8 weeks (a moving average) and the slow-mover methods Croston, SBA and TSB (reference slides)
- **Keep a simple benchmark in every comparison.**

# The Seasonal Naive Benchmark

Plush Bear, store: the four weeks from 1 December 2025, forecast with the sales up to the week of 24 November.



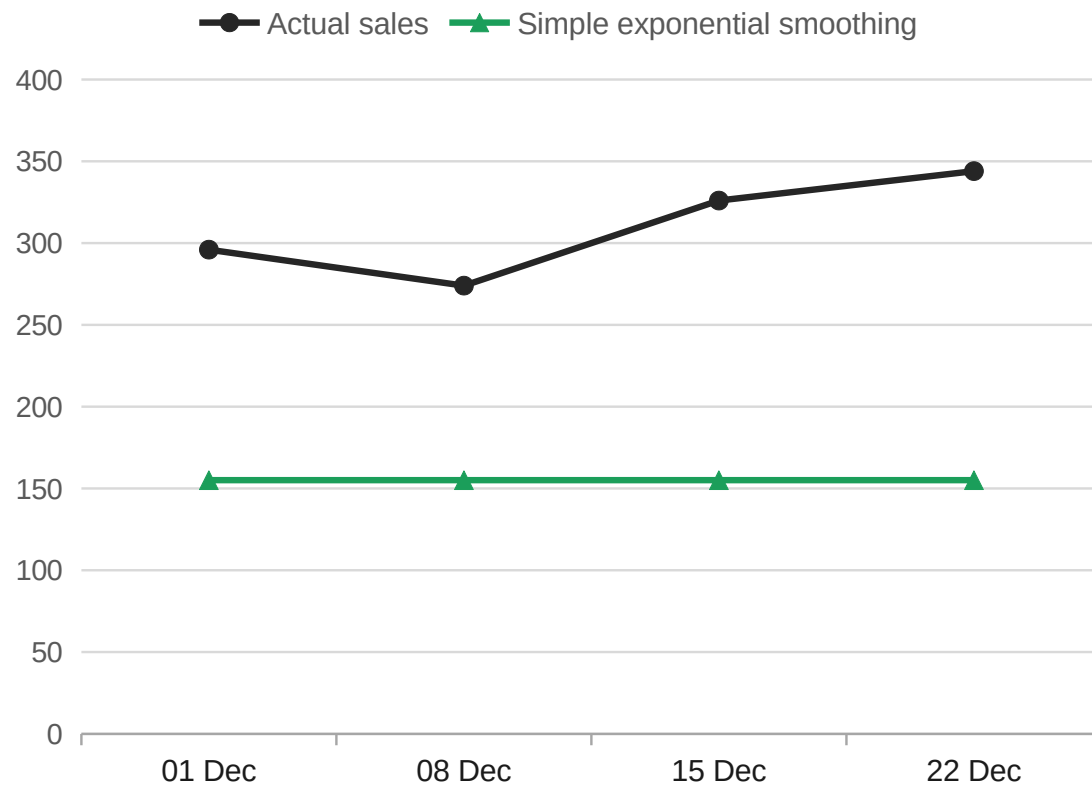
| Week   | Last year | Actual |
|--------|-----------|--------|
| 01 Dec | 228       | 296    |
| 08 Dec | 280       | 274    |
| 15 Dec | 324       | 326    |
| 22 Dec | 203       | 344    |

The forecast copies December 2024, week by week. The middle two weeks are almost exact (280 against 274, 324 against 326); the first and the last are far too low. The week of 22 December had a promotion in 2025 and none in 2024: the rule copies last year and cannot see this year's plan.

→ Seasonal naive gives us a clear benchmark to beat.

# Simple Exponential Smoothing

Same product, same forecast date, same four weeks: the smoothed level of the recent weeks.



**A flat 155 units a week:** the level on 24 November, carried forward. No new sales arrive for the December weeks, so nothing updates it, and it does not know that December sells more.

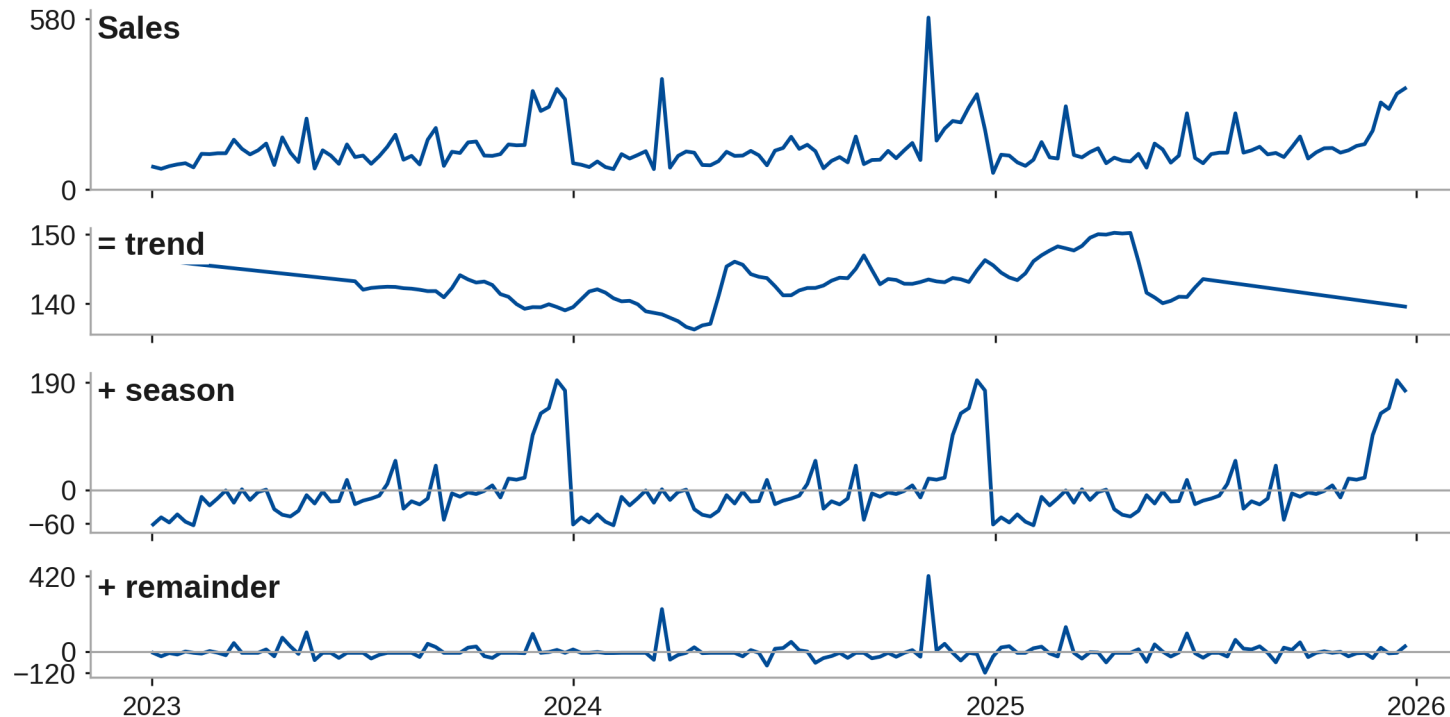
$$\begin{array}{ccccccc}
 \text{new level} & = & \alpha & \times & \text{this week's sales} & + & (1 - \alpha) \times \text{old level} \\
 \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 155 & \approx & 0.24 & \times & 200 & + & 0.76 \times 141 \\
 \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 \text{the forecast} & & \text{the fitting} & & \text{actual sales in} & & \text{1 - 0.24:} \\
 \text{for every} & & \text{algorithm's} & & \text{the week of} & & \text{the weight} \\
 \text{December week} & & \text{pick: fewest} & & \text{24 November, from} & & \text{of the past} \\
 & & \text{misses on past} & & \text{the data} & & \\
 & & \text{weeks} & & & & \\
 & & & & & & \text{not a sale:} \\
 & & & & & & \text{the smoothed} \\
 & & & & & & \text{level after} \\
 & & & & & & \text{17 November}
 \end{array}$$

*Top: the rule. Middle: the Bear's numbers on 24 November. Bottom: where each number comes from.*

→ Simple exponential smoothing follows the level, not the coming season.

# A Series In Three Parts

Time series decomposition: **sales = trend + season + remainder**, in every week. Plush Bear, store, 2023 to 2025.



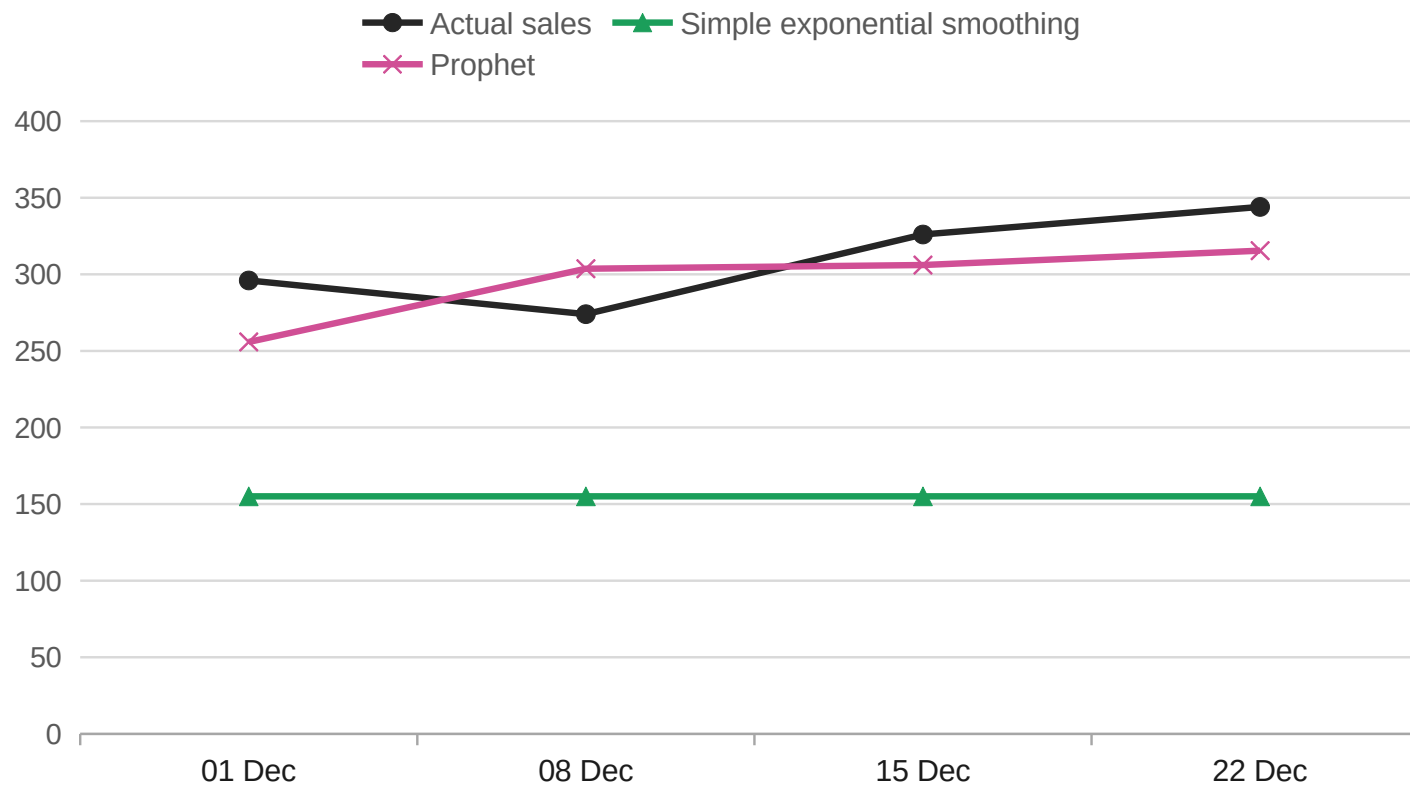
*Each panel has its own scale. The trend's first and last half-year are straight-line extensions; its steps in May 2024 and May 2025 are the 585-unit promotion week of November 2024 entering and leaving the 52-week window.*

→ **The same weeks can be read as parts that add up, or as a table of features.**

- **Trend:** the normal level, the average of the year around each week (a centred 52-week moving average). Here 136 to 150 units a week; it describes the past, not a forecast
  - **Season:** each week's typical distance from the trend, the same every year. The week before Christmas (week 51) adds 194 units
  - **Remainder:** what trend and season do not explain. In the 13 promotion weeks it averages +97 units, otherwise -9
- A statistician reads a series as parts. A machine-learning model reads it as features: a row of numbers for every week

# Prophet: Trend, Season And Promotions

A local model from Meta: one model per series, fitted on that series' own past. Plush Bear, store: the four weeks from 1 December 2025, forecast with the sales up to the week of 24 November.



| Prophet adds up  | What it is                                             |
|------------------|--------------------------------------------------------|
| Trend            | the level and its slope, allowed to bend a few times   |
| Yearly season    | the shape of a typical year, such as the December rise |
| Planned discount | the effect of this product's own promotion weeks       |

- On these four weeks it misses by **30 units** a week, simple exponential smoothing by 155: it sees December coming
- Whether it also wins across all 80 products and the whole year is the job of the backtest in chapter 3

→ A local model can see the season and the plan, but it learns from one product's past only.

# Local And Global Models

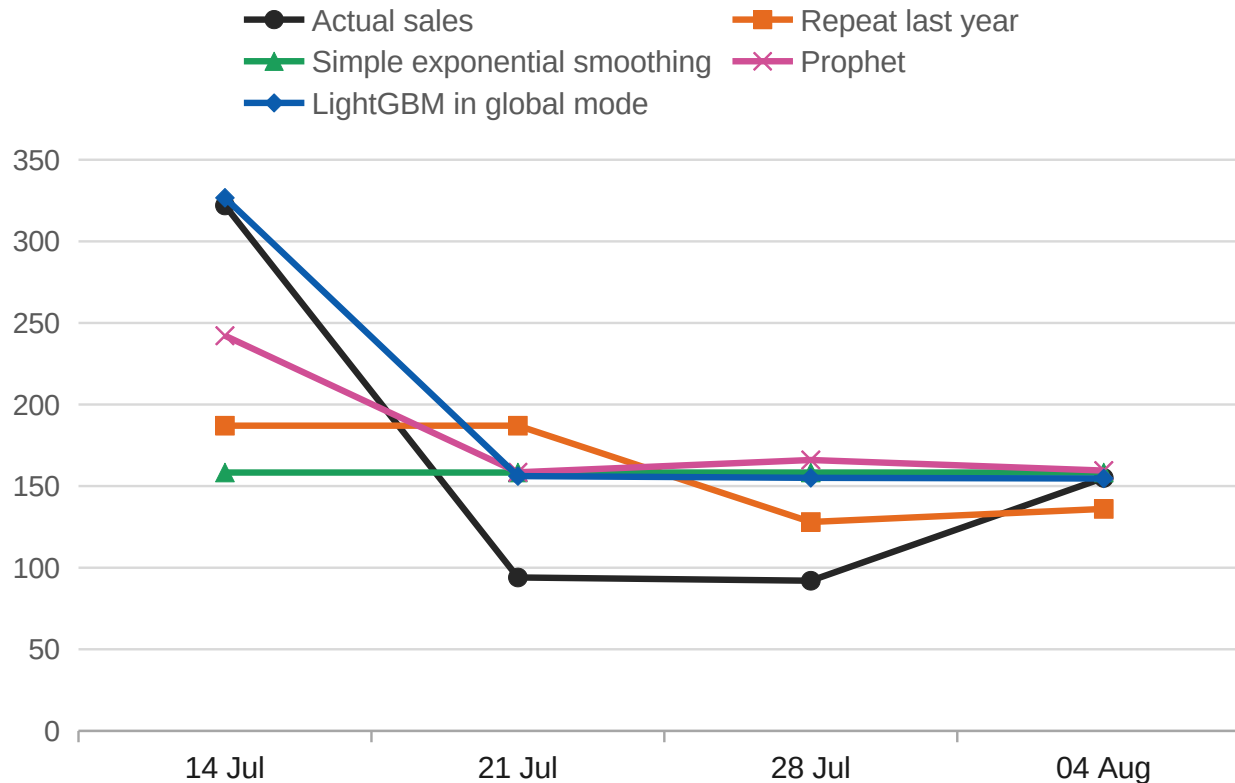
|                            | Local mode: one model per series (smoothing, Prophet, or LightGBM per series) | Global mode: one model for all series (our LightGBM)                                                            |
|----------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>What is fitted</b>      | Each series on its own history, about 2 to 3 years of weeks                   | One model on every row of every series at once                                                                  |
| <b>What it can learn</b>   | Only what its own series shows: level, trend, season                          | Patterns that repeat across products: a promotion appears 1,037 times in the data, about 13 times in one series |
| <b>New or short series</b> | Weak: a new product has no history                                            | Can borrow the season and the promotion effect from similar products (ours still needs a year of history)       |
| <b>Planned inputs</b>      | Possible, but each product learns the effect from its own few promotions      | Natural: one input column, one effect learned from all products                                                 |
| <b>Cost and risk</b>       | Fast, simple, easy to explain per series                                      | More preparation; the big products can dominate it; it needs a backtest before you trust it                     |

- **LightGBM**, a model built from many small decision trees, can run in either mode. We train it in **global mode**, so it learns rules such as "a 20 percent discount: expect about 2 times a normal week" from all products
- **Local or global is about the data a model learns from, not about the algorithm.**



# LightGBM In Global Mode At Work

Garden Hose, store: the four weeks from 14 July 2025, forecast with the sales up to 7 July. Week 1 had a planned 20 percent discount; the hose had seen only 6 promotions of its own.



**The promotion week: 322 sold.** LightGBM in global mode 327, Prophet 242, simple exponential smoothing 158, repeat last year 187. It learned what 20 percent off does from all 1,037 promotion weeks, not from the hose's 6.

| Average miss, units          | 349 promotion weeks | 3,809 normal weeks |
|------------------------------|---------------------|--------------------|
| Repeat last year             | 49.9                | 16.1               |
| Simple exponential smoothing | 45.4                | 12.3               |
| Prophet                      | 26.0                | 10.4               |
| LightGBM in global mode      | 23.7                | 9.7                |

*All 80 series, every week of 2025. Most of the gain comes from knowing the plan; global mode adds most for products with few promotions of their own (22.5 against 28.4 units for Prophet).*

→ **Global mode lets one product borrow what all the others have learned.**

# One Week Ahead Or Several

Single-step: forecast next week. Multi-step: forecast several weeks at once, which the stock chapter needs.

| Way to forecast several weeks                | How it works                                                              | Watch out                                                    |
|----------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Feed the forecast back in (recursive)</b> | Forecast next week, treat it as if it were sales, forecast the week after | Errors pile up: each step builds on the last guess           |
| <b>One model per week ahead (direct)</b>     | A separate model for 1, 2, 3 and more weeks ahead                         | Many models to build and to keep consistent                  |
| <b>One model, weeks ahead as a feature</b>   | One model learns every horizon at once: LightGBM in global mode           | A bigger training table: every week appears once per horizon |

| WAPE by weeks ahead                 | 1    | 2    | 3    | 4    | All 4       |
|-------------------------------------|------|------|------|------|-------------|
| <b>Repeat last year</b>             | 43 % | 43 % | 44 % | 45 % | <b>44 %</b> |
| <b>Simple exponential smoothing</b> | 34 % | 35 % | 35 % | 36 % | <b>35 %</b> |
| <b>Prophet</b>                      | 28 % | 27 % | 28 % | 26 % | <b>27 %</b> |
| <b>LightGBM in global mode</b>      | 25 % | 25 % | 26 % | 24 % | <b>25 %</b> |

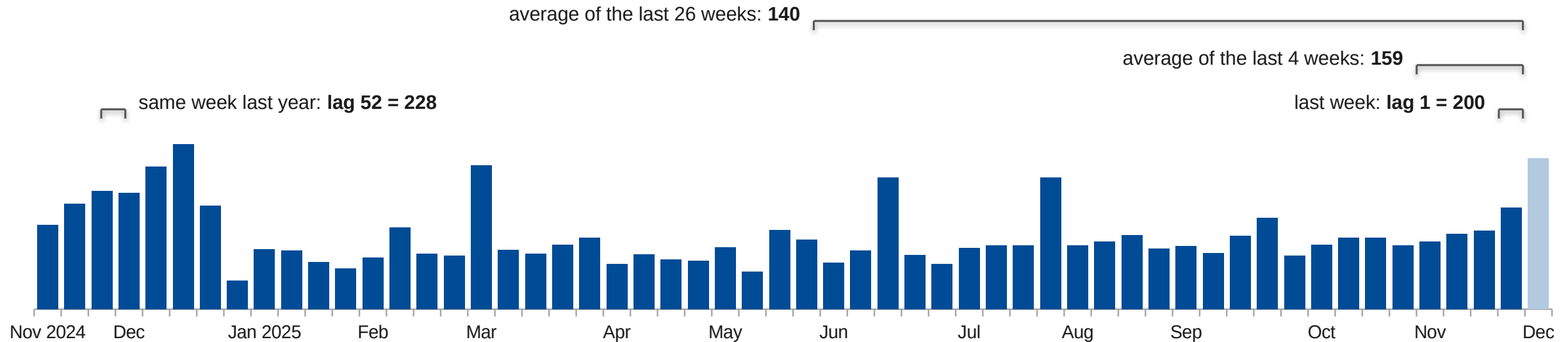
- The error changes by at most **3 points** from 1 to 4 weeks ahead: the calendar and the promotion plan are known in advance
- Further ahead, and for less regular products, errors usually grow. Test the horizon you order for

*One test, a replay of 2025: at 13 dates, every 4 weeks, each method forecast the next 4 weeks at once, for all 80 series. WAPE: total miss ÷ total sales.*

→ **An order covers several weeks, so the forecast must reach as far as the next delivery.**

# From A Series To Features

Plush Bear, store: the forecast made after the week of 24 November, the last week with known sales, for the week of 1 December 2025. The brackets mark where each number comes from.



*Light bar: the week we forecast. Its actual sales, 296 units, are the target the model learns to predict; they are unknown when the forecast is made.*

- **From the calendar:** week 49 of the year, the Christmas window. **From the plan:** promotion no, price EUR 25.00. **Identity:** Plush Bear, store, Toys

→ A feature is one number the model is given for a week; the model learns how they combine.

# One Row Of Features

All 26 features of one row: Plush Bear, week of 1 December 2025, one week ahead, for LightGBM in global mode trained on 2024. Weeks back count from the week being forecast.

| From the series itself                    | Value | Type   |
|-------------------------------------------|-------|--------|
| sales 1 week back                         | 200   | number |
| sales 2 weeks back                        | 154   | number |
| sales 3 weeks back                        | 148   | number |
| sales 4 weeks back                        | 133   | number |
| sales 8 weeks back                        | 126   | number |
| sales 12 weeks back                       | 110   | number |
| sales 26 weeks back                       | 91    | number |
| sales 52 weeks back (same week last year) | 228   | number |
| average of the last 4 weeks               | 159   | number |
| average of the last 8 weeks               | 146   | number |
| average of the last 13 weeks              | 141   | number |
| average of the last 26 weeks              | 140   | number |
| variation over the last 8 weeks           | 24.1  | number |

| History, calendar, plan, identity       | Value             | Type     |
|-----------------------------------------|-------------------|----------|
| share of zero weeks, last 13            | 0 %               | number   |
| weeks since the last sale               | 0                 | number   |
| week of the year                        | 49                | number   |
| month                                   | 12                | number   |
| Christmas window (weeks 48 to 51)       | yes               | yes/no   |
| Black Friday week (week 47 in our data) | no                | yes/no   |
| promotion planned                       | no                | yes/no   |
| discount planned                        | 0 %               | number   |
| price planned                           | EUR 25.00         | number   |
| which product                           | P010 (Plush Bear) | category |
| which channel                           | store             | category |
| which category                          | Toys              | category |
| weeks ahead                             | 1                 | number   |

→ The model was trained on 28,000 rows like this one: all 80 series, the weeks of 2024, each for 1 to 7 weeks ahead.

# Internal And External Features

Internal features come from the company's own data, external ones from outside it. Everything must be known on the day of the forecast.

| Where it comes from                         | Typical features                                                                               | Known in advance?            | In our model                         |
|---------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------|
| <b>Internal: the product's own history</b>  | last weeks' sales, averages over 4 to 26 weeks, weeks since the last sale, share of zero weeks | Yes, by construction         | Yes, 15 features                     |
| <b>External: the calendar</b>               | week of the year, month, public and school holidays, Black Friday, paydays                     | Yes                          | Week, month, Black Friday, Christmas |
| <b>Internal: planned commercial actions</b> | promotions, discount depth, price, in-store display, advertising, a new listing                | Yes, if it is planned        | Promotion, discount, price           |
| <b>Internal: product and place</b>          | category, price band, pack size, lifecycle stage, store size, region, channel                  | Yes                          | Product, category, channel           |
| <b>External: outside the company</b>        | weather, local events, competitor promotions, economic indicators                              | Only as forecasts themselves | No                                   |
| <b>Internal: the stock situation</b>        | stock-outs, shelf availability                                                                 | Not for future weeks         | No                                   |

*Plus one more feature: how many weeks ahead the forecast is (1 to 7). Together the 26 features of slide 28.*

→ **Only information available when the forecast is made may be used.**

# Choosing The Features

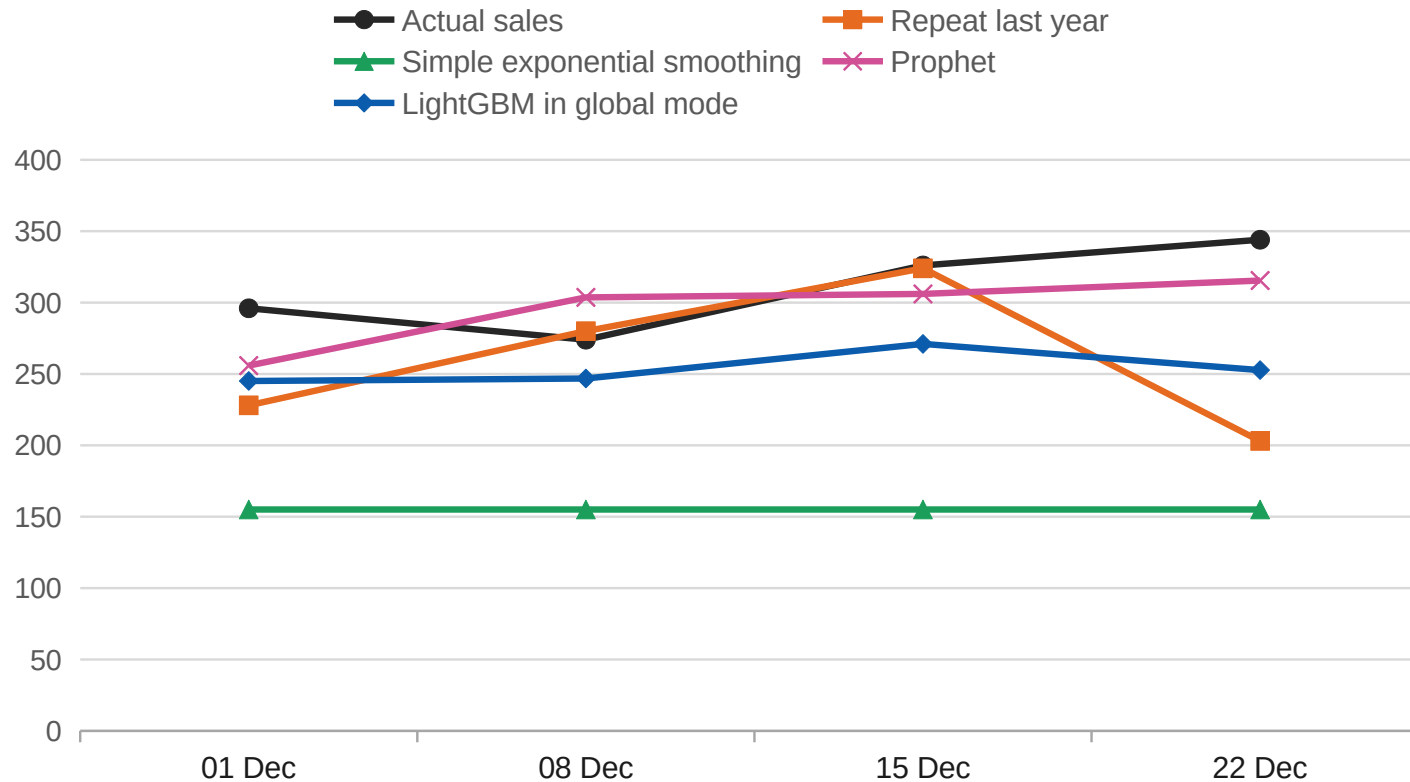
Feature selection: more features is not better. Each one has to earn its place.

- **Leakage.** A feature that contains information from the future, such as next week's stock-out flag or a price that was changed after the forecast was made, makes the test look very good and the real forecast fail
- **Noise.** A feature that carries no information about demand makes the model a little worse and much harder to explain. Weather at weekly product level is a common example
- **Cost.** Every feature has to be delivered every week, for every product, for years. A feature nobody maintains is a forecast that stops working
- **How to decide.** Start from what a planner would look at: history, promotions, calendar. Add one feature at a time and keep it only if the backtest error falls
- **How to check.** Scramble one feature and see how much worse the forecast gets. Scrambling the average of the last 26 weeks raises the error (WAPE) from 25 % to 68 %; scrambling the product identity raises it to 65 %. Most of the other 24 features change it by less than 1 point

→ A short list the business can explain and supply beats a long list the model hardly uses.

# The Same Four Weeks Compared

Plush Bear, store, the same four weeks, all four approaches. Average miss: how far off per week, in units. Bias: forecast minus actual on average.



| Approach                     | Average miss | Bias               |
|------------------------------|--------------|--------------------|
| Repeat last year             | 54 units     | -51 units (-17 %)  |
| Simple exponential smoothing | 155 units    | -155 units (-50 %) |
| Prophet                      | 30 units     | -15 units (-5 %)   |
| LightGBM in global mode      | 56 units     | -56 units (-18 %)  |

All four forecast too low this December. Prophet misses least, 30 units a week; simple exponential smoothing misses by 155. Would you choose the same for the summer, or for the online channel?

→ One product and one window are a starting point, not a verdict.

# Models Trained On Many Datasets

Foundation models: trained once on millions of time series, then used on your products without any training on them (zero-shot).

| Model            | Who        | What is special                                                                                                                                   |
|------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TimesFM</b>   | Google     | One of the first; trained on a very large collection of time series                                                                               |
| <b>Chronos</b>   | Amazon     | Treats a time series like a sentence of words                                                                                                     |
| <b>Moirai</b>    | Salesforce | Handles daily, weekly and monthly data and extra features in one model                                                                            |
| <b>TabPFN-TS</b> | Prior Labs | A general model for tables, adapted to forecasting                                                                                                |
| <b>TiRex</b>     | NXAI, Linz | Built on xLSTM, a neural network for sequences, from Sepp Hochreiter's group with JKU's LIT AI Lab; first in public zero-shot comparisons in 2025 |

- The extreme case of a global model: trained once on everything, no fitting on your data, and it returns a range, not only one number
- What to watch: most cannot use promotion or price features, slow movers are not their strength, and they must pass the same backtest as every other method

→ **Treat a new model as another candidate for the same test.**



# Choosing A Method

| Situation                                    | Start with                                                         | Then try                                                                                        | Avoid                                                                                      |
|----------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Smooth, seasonal, 2+ years of history</b> | Seasonal naive as the benchmark; SES                               | Prophet with the planned discount; LightGBM in global mode with promotion and calendar features | Choosing a method without a backtest                                                       |
| <b>Smooth, 1 to 2 years of history</b>       | SES, or the average of the last 8 weeks (a moving average)         | LightGBM in global mode, once the product has a year of history                                 | Holt-Winters (too little history to learn the season)                                      |
| <b>Erratic</b>                               | SES or a moving average                                            | LightGBM in global mode                                                                         | Seasonal naive alone (it copies last year's random ups and downs); Prophet overreacts here |
| <b>Intermittent</b>                          | TSB or Croston SBA                                                 | LightGBM in global mode; judge the total over the lead time                                     | Holt-Winters; choosing by a weekly percentage error (MAPE, chapter 3)                      |
| <b>Lumpy</b>                                 | TSB or Croston SBA for the rate; plan the total over the lead time | LightGBM in global mode; compare by the stock outcome                                           | Any choice based on weekly accuracy                                                        |
| <b>Strong promotion or price effects</b>     | LightGBM in global mode with planned features                      | Prophet with the discount as an input, product by product                                       | Local methods without the promotion plan                                                   |
| <b>New product</b>                           | Similar products, the category average, judgement                  | A global model built for it (ours needs a year of history)                                      | Any method that needs the product's own history                                            |

→ **Start simple, keep the benchmark, and let the backtest decide.**

# Hands-On 1: Choosing A Forecast

Open **Notebook 1**, sections 1 to 3. Copy to Drive, Run all, work in pairs.

- Section 1: read the channel picture and the four example products
- Section 2: the Christmas comparison for the Plush Bear. Then change the window to Summer, and the product to Garden Hose
- Try Plush Bear in store and online. Does your choice change with the channel?
- Section 3: write a choice, one supporting number and one limitation, and explain it to your partner
- Stuck? Runtime, Restart session and run all. Still stuck? Raise a hand, that is what I am here for

→ **Minimum: one comparison, and explain your choice to your partner.**

# Evaluate

Metrics, honest tests, tuning, and  
the model's reasons

03

# The Size Of The Miss: MAE And WAPE

A teaching example with round numbers: four weeks, one forecast.  $|x|$  means the size of  $x$  without its sign.

| Week  | Actual | Forecast | Miss = forecast - actual | Miss |
|-------|--------|----------|--------------------------|------|
| 1     | 100    | 90       | -10                      | 10   |
| 2     | 120    | 100      | -20                      | 20   |
| 3     | 80     | 70       | -10                      | 10   |
| 4     | 100    | 100      | 0                        | 0    |
| Total | 400    | 360      | -40                      | 40   |

## MAE, mean absolute error

- In words: add up the sizes of the misses and divide by the number of weeks
- $40 \div 4 = \mathbf{10 \text{ units}}$  a week

## WAPE, weighted absolute percentage error

- In words: total size of the misses divided by total demand
- $40 \div 400 = \mathbf{10 \%}$ . Comparable across products of different size

→ Take the size of each miss before averaging, so highs and lows do not cancel.

# Accuracy And Direction Together

Two forecasts with the same total miss of 40 units on the same 400 units of demand.

|                                                      | Forecast A            | Forecast B                       |
|------------------------------------------------------|-----------------------|----------------------------------|
| Weekly forecasts (actual 100, 120, 80, 100)          | 90, 100, 70, 100      | 110, 100, 90, 100                |
| Total  miss                                          | 40 units              | 40 units                         |
| WAPE: total  miss  ÷ total demand                    | 10 %                  | 10 %                             |
| Bias: (total forecast - total demand) ÷ total demand | -10 %: never too high | 0 %: too high and too low cancel |

- **Bias** in words: the average direction of the miss. Positive means the forecast runs too high, negative too low
- A stock rule fed by forecast A runs out; fed by forecast B it holds too much in some weeks and too little in others. Same accuracy number, different shelves

→ Same WAPE does not mean the same inventory consequences.

# Accuracy On A Slow Mover

A slow mover, with round numbers: one sale in four weeks, and a forecast that always says zero.

|                | Week 1 | Week 2 | Week 3 | Week 4 |
|----------------|--------|--------|--------|--------|
| Actual units   | 0      | 0      | 0      | 4      |
| Forecast units | 0      | 0      | 0      | 0      |
| Correct?       | yes    | yes    | yes    | no     |

## MAPE, mean absolute percentage error

- In words: each week's  $|\text{miss}|$  divided by that week's demand, averaged over the weeks
- Dividing by zero demand: **undefined** in three of four weeks

## WAPE

- Total  $|\text{miss}|$  4  $\div$  total demand 4 = **100 %**: every unit of demand missed
- Three correct weeks and an empty shelf. For products with many zero weeks, judge the stock decision, not the share of correct weeks

→ Judge a forecast by the demand it covers, not by how many weeks it was right.

# The Metrics In Words

Five numbers you will meet in every forecast report.  $|x|$  is the size of  $x$  without its sign. Each metric is blind somewhere.

| Metric                                           | How it is computed                                                                                       | In words                                                                             | Blind when                                                                              | Use it for                                       |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>MAE (mean absolute error)</b>                 | average of $ \text{actual} - \text{forecast} $                                                           | the typical miss, in units                                                           | products of different size are compared                                                 | one product, reported to its planner             |
| <b>Bias</b>                                      | $(\text{total forecast} - \text{total actual}) \div \text{total actual}$                                 | the average direction of the miss; positive means too high                           | on its own: it says nothing about accuracy                                              | every table, next to the accuracy number         |
| <b>MAPE (mean absolute percentage error)</b>     | average of $ \text{actual} - \text{forecast}  \div \text{actual}$                                        | the typical miss as a percentage of that week's demand                               | any week with zero demand; weeks with low demand; it favours forecasts that are too low | a single fast mover, if at all                   |
| <b>WAPE (weighted absolute percentage error)</b> | $\text{sum of }  \text{actual} - \text{forecast}  \div \text{sum of actual}$                             | the total miss as a percentage of total demand                                       | small products disappear behind big ones                                                | a category or an assortment: the working default |
| <b>MASE (mean absolute scaled error)</b>         | $\text{MAE} \div \text{MAE of the naive forecast (each week = the week before) on the training history}$ | the miss compared with the past miss of simply repeating last week; below 1 beats it | a flat training history; fine with zero weeks                                           | comparing methods across all kinds of products   |

→ **Never accept an accuracy number without the bias next to it.**

# Backtest: Replaying The Past

A backtest replays the past: stand at a date, forecast the next weeks with what was known then, compare, move on.

| Step                              | What we do                                                                                                       | In our backtest                                                 |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>1. Stand at a past date</b>    | Pick a forecast date (the origin: the last week with known sales); train every method on the weeks up to it only | Origins every 4 weeks from 23 December 2024 to 24 November 2025 |
| <b>2. Forecast the next weeks</b> | Forecast the 4 weeks after it (the test window) with every method, and score them                                | 4 weeks ahead, all 80 series, flagged weeks left out            |
| <b>3. Move forward and repeat</b> | Collect the misses over the whole year                                                                           | 13 test dates, 4,160 forecasts per method, 4,158 scored         |

- Never assign weeks at random to training and test: the model would have seen the weeks right before and after, and the test is too easy
- One test window is one sample: the four weeks before Christmas rank the methods differently from four weeks in June

→ **Ask: which dates, how far ahead, and what was known at the time?**



# Backtest Schemes

| Scheme                | How                                                                                             | Good for                                                      | Weakness                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Fixed split</b>    | One date; train before, test after                                                              | A first look                                                  | One test window: Christmas ranks methods differently from June                             |
| <b>Rolling origin</b> | Many dates; at each one, train on everything before and test on the weeks after                 | The standard; it works the way forecasts are used in practice | More work: every method is retrained at every date                                         |
| <b>Sliding window</b> | Like rolling origin, but train only on a fixed number of recent weeks, for example the last 104 | A business that changed                                       | Forgets the history the season needs                                                       |
| <b>Random split</b>   | Weeks assigned at random to training or test                                                    | Never for time series                                         | The model has seen the weeks right before and after the test week, so the test is too easy |

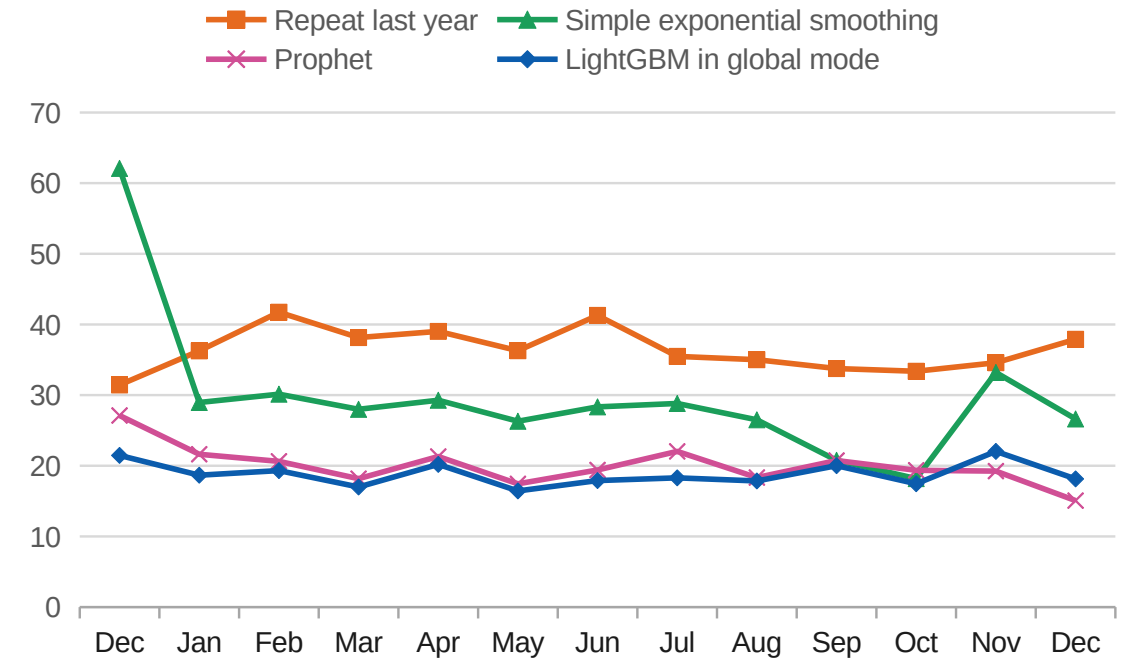
- Ours is a rolling origin: 13 test dates, 4 weeks ahead, every method retrained at every date, flagged weeks left out of the scoring

→ **When someone shows you a comparison of methods, ask two things: how many test dates, and which weeks of the year.**

# Fixed Split And Rolling Origin

Smooth series: the same methods judged on one test window (the four weeks from 1 December 2025) and on all 13 windows.

| Method                       | One window | Rank | All windows | Rank |
|------------------------------|------------|------|-------------|------|
| LightGBM in global mode      | 18.1 %     | 2    | 18.8 %      | 1    |
| Prophet                      | 15.1 %     | 1    | 19.8 %      | 2    |
| Croston SBA                  | 34.1 %     | 7    | 29.2 %      | 3    |
| Simple exponential smoothing | 26.6 %     | 3    | 29.4 %      | 4    |
| TSB                          | 33.2 %     | 6    | 30.3 %      | 5    |
| Moving average (8 weeks)     | 31.5 %     | 4    | 31.6 %      | 6    |
| Holt-Winters                 | 33.0 %     | 5    | 32.4 %      | 7    |
| Repeat last year             | 37.9 %     | 8    | 36.6 %      | 8    |



- Prophet is first on the four weeks before Christmas, LightGBM in global mode over the year. Croston SBA moves from rank 7 to rank 3; only Repeat last year stays in its place, last

*WAPE per test window in percent, smooth series, labelled by the month of the first test week (30 Dec 2024 to 1 Dec 2025).*

→ One test window is one sample: Prophet wins the four weeks before Christmas, LightGBM in global mode wins the year.

# Backtest Results Across The Year

WAPE and bias over all 80 series and 13 test dates, then WAPE by kind of demand. Bold marks the lowest WAPE in each column.

| Method                       | WAPE, all     | Bias, all | WAPE Smooth | WAPE Erratic | WAPE Intermittent | WAPE Lumpy   |
|------------------------------|---------------|-----------|-------------|--------------|-------------------|--------------|
| Repeat last year             | 43.8 %        | -2.7 %    | 37 %        | 95 %         | 145 %             | 194 %        |
| Moving average (8 weeks)     | 37.5 %        | +0.2 %    | 32 %        | 79 %         | 135 %             | 162 %        |
| Simple exponential smoothing | 34.9 %        | -0.9 %    | 29 %        | 74 %         | 129 %             | 160 %        |
| Holt-Winters                 | 39.0 %        | -1.0 %    | 32 %        | 86 %         | 143 %             | 189 %        |
| Croston SBA                  | 34.9 %        | -4.9 %    | 29 %        | 76 %         | 128 %             | 158 %        |
| TSB                          | 36.1 %        | +0.0 %    | 30 %        | 77 %         | 132 %             | 162 %        |
| Prophet                      | 27.1 %        | +0.3 %    | 20 %        | 79 %         | 131 %             | 173 %        |
| LightGBM in global mode      | <b>25.1 %</b> | -3.1 %    | <b>19 %</b> | <b>71 %</b>  | <b>121 %</b>      | <b>137 %</b> |

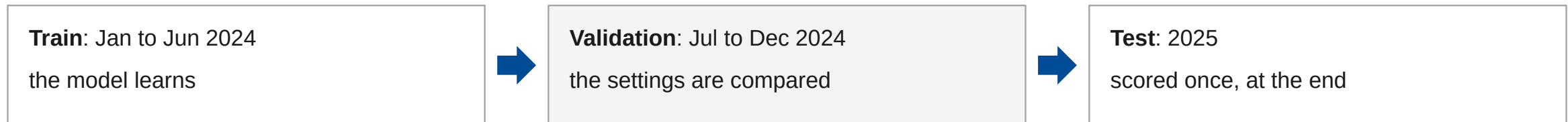
- LightGBM in global mode has the lowest WAPE in every column, but runs low wherever sales are irregular (-16 % erratic, -15 % intermittent, -35 % lumpy)
- Prophet comes next (27.1 %) and is almost unbiased, but on erratic, intermittent and lumpy series it does worse than simple exponential smoothing

→ An assortment average gives large sellers more influence; read the pattern columns and the bias too.

# Tuning The Model's Settings

Parameters are learned from the data. Hyperparameters are the settings chosen before training; tuning searches for good ones.

| Model                          | Learned from the data (parameters)                                    | Chosen before training (hyperparameters)                                                                                                                                                                                                  |
|--------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exponential smoothing</b>   | the level, and how much the newest week counts (the smoothing weight) | trend or not, damping or not, season length of 52 weeks                                                                                                                                                                                   |
| <b>LightGBM in global mode</b> | every decision rule (split) and end value (leaf) of every tree        | number of trees; learning rate (how much each new tree corrects the ones before); tree size (leaves per tree, rows per leaf); share of rows and of features each tree sees; Tweedie power (how it treats weeks with zero and small sales) |

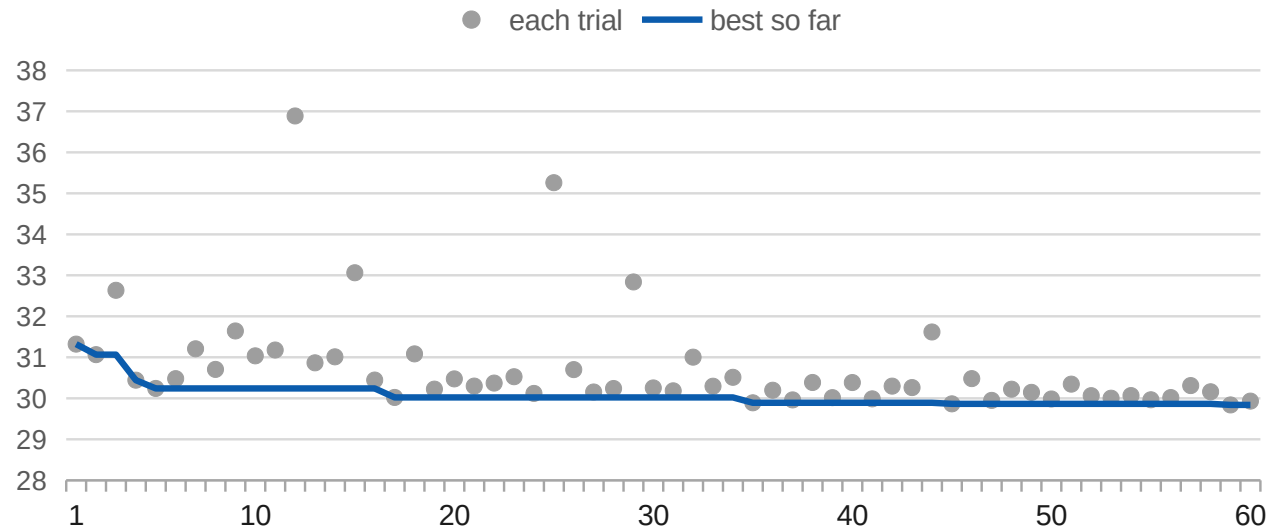


- **The goal:** the settings with the smallest error on weeks the model did not learn from
- **The search:** a grid tries every combination, a random search tries a sample, and Optuna, a free Python library, learns from each finished trial (one training run with one setting) where to look next

→ **Tune on the validation weeks; the test year is used once, at the end.**

# Tuning With Optuna

60 trials: each trains LightGBM in global mode with one set of settings on Jan to Jun 2024 and scores it on Jul to Dec 2024 (WAPE, 1 to 4 weeks ahead).



Validation WAPE per trial, percent; trial 1 is our own setting. The 2025 row trains once, without retraining (the backtest on slide 43: 25.1 %).

| Setting                         | Our setting   | Tuned         |
|---------------------------------|---------------|---------------|
| learning rate                   | 0.030         | 0.022         |
| number of trees                 | 600           | 350           |
| leaves per tree                 | 31            | 27            |
| minimum rows per leaf           | 20            | 15            |
| share of rows per tree          | 0.80          | 0.92          |
| share of features per tree      | 0.80          | 0.43          |
| Tweedie power                   | 1.30          | 1.65          |
| <b>WAPE, validation weeks</b>   | <b>31.3 %</b> | <b>29.8 %</b> |
| <b>WAPE, 2025, one training</b> | <b>25.5 %</b> | <b>25.6 %</b> |

**Vibe-code it yourself: describe the task in plain words and let Gemini in Colab write the code**

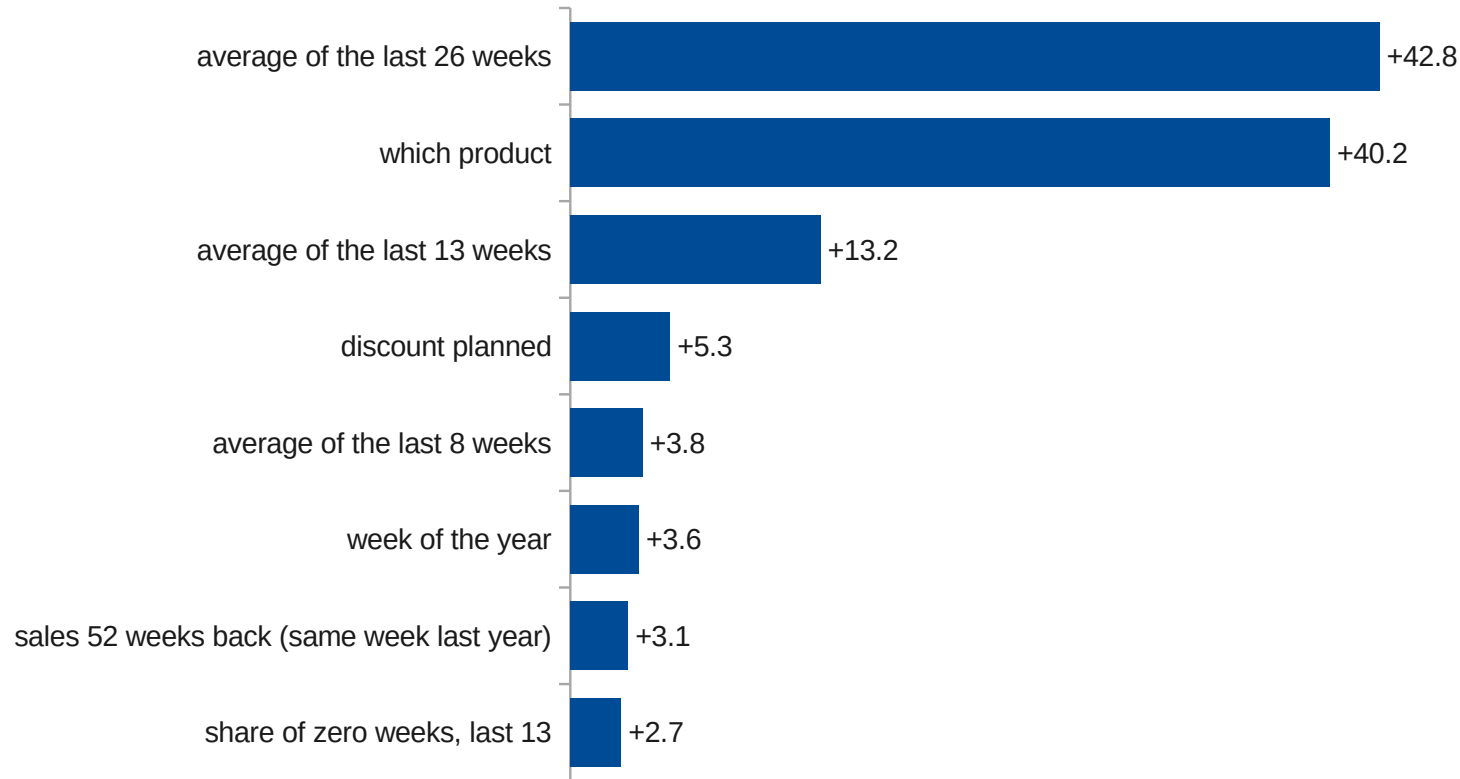
Write Python code that tunes a LightGBM forecasting model with Optuna. Train on January to June 2024 (2023 only feeds the lag features), score every trial by WAPE on July to December 2024, start from learning rate 0.03, 600 trees, 31 leaves, 20 rows per leaf, 0.8 row and feature shares and Tweedie power 1.3, run 60 trials, never use data from 2025, and print the best settings next to these.

→ **Tuning lowered the validation WAPE by 1.5 points and did not help on the test year: our settings were already good.**

# What The Model Relies On

Permutation importance: scramble one feature in the 2025 data and see how much worse the forecast gets.

Without scrambling: WAPE 25.1 %.



*The 8 largest of 26 features. Increase in WAPE, percentage points, average of 3 scrambles.*

→ **These are the model's habits, not the causes of demand.**

- The model relies most on the product's **recent average sales** and on **which product** it is: scramble either and the error rises from 25 % to 65 to 68 %, more than two and a half times
  - **Discount** and **promotion** rank lower because they are on in about one week in 12; when they are on, they change the number a lot
  - Of 26 features, most cost under one point. They could be removed, and the model would be easier to explain
- A short, explainable list of features is a strength, not a weakness

# One Forecast Explained

Plush Bear, store, week of 1 December 2025, one week ahead, from LightGBM in global mode, trained once on data up to 2024: forecast 274, actual 296.

| Feature                                   | Its value  | Factor |
|-------------------------------------------|------------|--------|
| which product                             | Plush Bear | x 4.16 |
| average of the last 26 weeks              | 140        | x 3.99 |
| average of the last 13 weeks              | 141        | x 1.34 |
| week of the year                          | 49         | x 1.22 |
| sales 52 weeks back (same week last year) | 228        | x 1.12 |
| average of the last 8 weeks               | 146        | x 1.09 |
| all other features together               |            | x 1.34 |

Start from the model's **reference level of 6.17 units** (its starting value: a typical week across all 80 series, most of them small sellers) and multiply by every factor: 274 units. No promotion this week: the promotion and discount factors are close to 1 and sit in the last row.

**In a meeting:** "We forecast 274. The bear's recent weeks and the December week explain most of the number; actual demand was 296, so our forecast was 22 units too low."

→ A factor tells you how the model built the number, not why customers bought.

# From Forecast To Stock

The protection period, safety stock,  
one order, and what a service  
target buys

04



# From A Weekly Forecast To Stock

| Term                     | Plain meaning                                                                     | Our example |
|--------------------------|-----------------------------------------------------------------------------------|-------------|
| <b>Lead time</b>         | The wait between placing an order and receiving it                                | 4 weeks     |
| <b>Review period</b>     | How often we look at the stock and place an order                                 | every week  |
| <b>Protection period</b> | Lead time plus review period: the weeks the stock on hand and on order must cover | 5 weeks     |

- Whatever we order today arrives in four weeks and has to last until the next delivery, one week later. So today's decision must cover five weeks of demand
- The next five slides build one order from that: add the forecasts, add a cushion, see where its size comes from, subtract what is already there

→ **First decide which weeks your stock must cover.**

# Forecast Over The Protection Period

Round teaching numbers: five weekly forecasts for the five weeks the stock must cover.

| Week ahead      | 1   | 2   | 3   | 4   | 5   | Sum   |
|-----------------|-----|-----|-----|-----|-----|-------|
| Forecast, units | 180 | 190 | 210 | 200 | 220 | 1,000 |

**Forecast over the protection period: 1,000 units.**

- If the forecast were perfect, 1,000 units on hand and on order would be exactly enough. It is not perfect, so we add a cushion next

→ **Use the sum over the protection period, not one week's value.**

# Safety Stock

The cushion is called safety stock. It is extra stock, on top of the forecast.

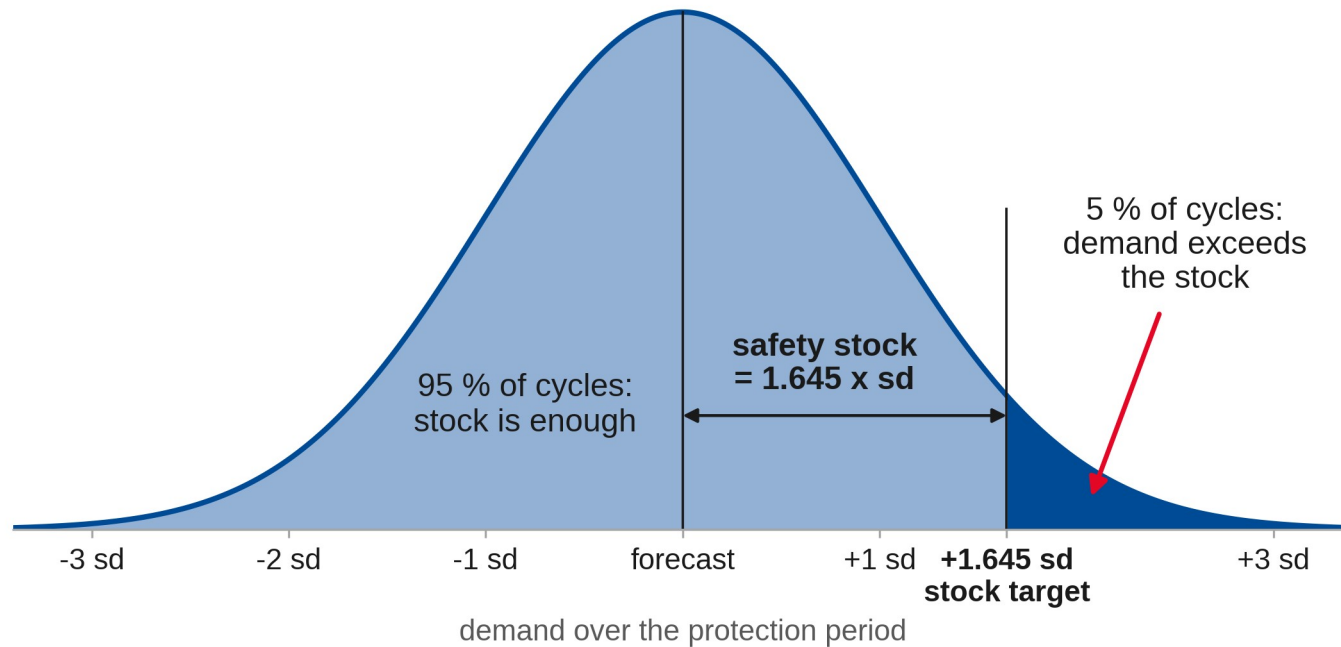
| Quantity                                     | Meaning                                                                                                                                    | Value            |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Error spread (standard deviation, sd)</b> | How widely the weekly misses scatter around their average, from the backtest; for a bell curve about 1.25 x the average miss (MAE)         | 40 units         |
| <b>Cycle service target</b>                  | The share of order cycles (here one week, from one delivery to the next) that we plan to get through without a stock-out                   | 95 %             |
| <b>Protection period</b>                     | The weeks the cushion has to cover                                                                                                         | 5 weeks          |
| <b>Safety stock</b>                          | <b>z for the target x error spread x square root of the weeks: <math>1.645 \times 40 \times \sqrt{5} = 147.1</math>, rounded up to 148</b> | <b>148 units</b> |

- Why the square root: over five weeks the misses partly cancel out, so the spread grows by about 2.2 times, not 5 times. That assumes the weekly misses are independent
- Where does 1.645 come from? From the bell curve, next slide

→ **Safety stock depends on how wrong the forecast tends to be, and on your service target.**

# The Safety Factor z

If the misses follow a bell curve, demand stays below the forecast plus z standard deviations (sd) in a fixed share of cycles. For a 95 percent target,  $z = 1.645$ .



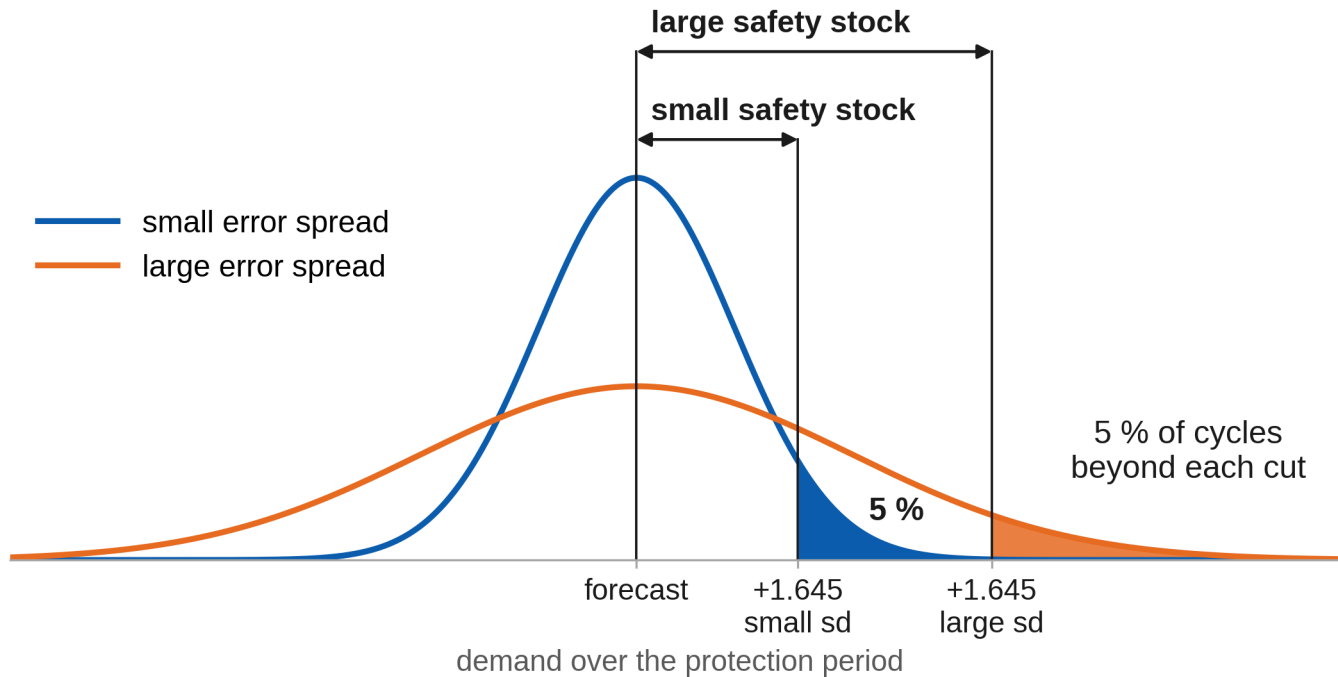
| Target      | z            | Safety stock, our example |
|-------------|--------------|---------------------------|
| 50 %        | 0.00         | 0 units                   |
| 84 %        | 1.00         | 90 units                  |
| 90 %        | 1.28         | 115 units                 |
| <b>95 %</b> | <b>1.645</b> | <b>148 units</b>          |
| 99 %        | 2.33         | 209 units                 |

- Spread over the 5 weeks:  $40 \times \sqrt{5} = 89.4$  units. Safety stock =  $z \times 89.4$ , rounded up to whole units
- From 95 to 99 percent the cushion grows by 41 percent for four points of service: the curve is thin at the edge

→ z belongs to the bell curve; the spread is yours, from the backtest; the target is the business's choice.

# One z For Every Product

Every bell curve has the same shape. Measured in standard deviations (sd), 95 percent of cycles always stay below +1.645 sd.



| Store, 95 % target | z     | Spread over L + 1 weeks | Safety stock |
|--------------------|-------|-------------------------|--------------|
| Plush Bear         | 1.645 | 88.8 units              | 147 units    |
| Puzzle 1000        | 1.645 | 67.3 units              | 111 units    |
| Vacuum Filter      | 1.645 | 2.45 units              | 5 units      |

- The same z for every product: the cut sits 1.645 spreads above the forecast, and 5 percent of cycles lie beyond it
- What differs is the spread in units, so the safety stock differs: 147 units for the Bear, 5 for the Filter
- This holds if the misses follow a bell curve; the replay later in this chapter checks it

→ z counts standard deviations, not units: one table serves every product.

# Order Quantity

The stock target (order-up-to level) is what we want on hand and on order together. We order only the difference.

| Quantity                                                                 | Units                             |
|--------------------------------------------------------------------------|-----------------------------------|
| <b>Stock target = forecast over the protection period + safety stock</b> | $1,000 + 148 = 1,148$             |
| <b>Already on the shelf + already on order</b>                           | $300 + 500 = 800$                 |
| <b>Order now = stock target - what is already there</b>                  | $1,148 - 800 = 348 \text{ units}$ |

- Ordering the whole target would count the 800 units twice. Every week the same rule runs again with the new forecast and the new stock on the shelf and on order
- Assumptions worth saying aloud: fixed lead time, deliveries as scheduled, no backorders (a customer who finds the shelf empty does not wait: the sale is lost), and misses that follow a bell curve

→ **Count what is already on the shelf and already on order.**

# Safety Stock For Three Products

Orders every week; protection period = lead time L plus one review week; safety stock =  $z \times \text{weekly spread (sd)} \times \sqrt{L + 1}$ , rounded up.

| Product              | Lead time | Units / week,<br>Jul to Dec 2025 | Weekly spread<br>(sd) | Spread over<br>L + 1 weeks | Safety<br>stock at<br>90 % | Safety<br>stock at<br>95 % | Safety<br>stock at<br>99 % | Safety stock in weeks<br>of demand, 95 % |
|----------------------|-----------|----------------------------------|-----------------------|----------------------------|----------------------------|----------------------------|----------------------------|------------------------------------------|
| Plush Bear, store    | 4 weeks   | 165.8                            | 39.7                  | 88.8                       | 114                        | 147                        | 207                        | 0.9                                      |
| Puzzle 1000, store   | 5 weeks   | 33.9                             | 27.5                  | 67.3                       | 87                         | 111                        | 157                        | 3.3                                      |
| Vacuum Filter, store | 6 weeks   | 0.58                             | 0.92                  | 2.45                       | 4                          | 5                          | 6                          | 8.7                                      |

*Weekly spread measured on January to June 2025, units per week over the 26 replay weeks.*

- The Puzzle sells 20 percent of the Bear's volume but needs 76 percent of its safety stock: its weekly spread is almost as large as its sales
- For the Filter the safety stock is 8.7 weeks of demand. The z factors assume that the misses follow a bell curve. For a product that sells single units, this assumption is poor. The replay on the next slides is the better check
- $z = 1.28, 1.645$  and  $2.33$  for 90, 95 and 99 percent. Each step up the curve buys less service for more stock

→ **Every unit of error spread you remove saves  $z \times$  the square root of the protection period in safety stock.**

# Service Target Versus Fill Rate

| Measure                           | The question it answers                                                                                   | Target or result?   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------|
| <b>Cycle service target</b>       | How likely do we want each order cycle (one week, from one order to the next) to end without a stock-out? | A target we choose  |
| <b>Fill rate</b>                  | What share of the units customers asked for did we actually serve?                                        | A result we measure |
| <b>Average stock on the shelf</b> | How much stock did the rule hold, on average?                                                             | A result we measure |
| <b>Weeks with a stock-out</b>     | How often did demand exceed the stock available?                                                          | A result we measure |

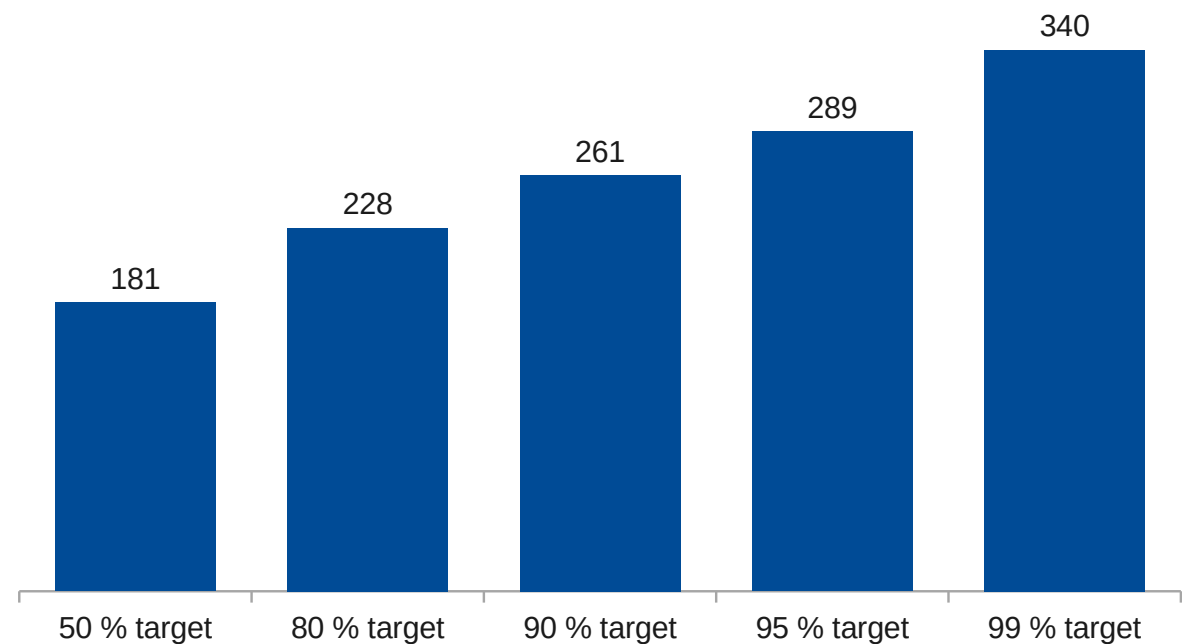
- How we measure the fill rate: replay the ordering rule week by week over 26 weeks; each week, served = the smaller of demand and stock on hand; fill rate = total units served ÷ total units demanded over the 26 weeks

→ A 95 percent cycle service target does not mean 95 percent of units served.



# Effect Of A Higher Service Target

Plush Bear, store: the ordering rule replayed week by week, 2025-06-30 to 2025-12-22, at five service targets.



Average units on the shelf.

| Target | Fill rate | Stock-out weeks | Units not served |
|--------|-----------|-----------------|------------------|
| 50 %   | 98.0 %    | 7               | 85               |
| 80 %   | 100.0 %   | 0               | 0                |
| 90 %   | 100.0 %   | 0               | 0                |
| 95 %   | 100.0 %   | 0               | 0                |
| 99 %   | 100.0 %   | 0               | 0                |

From 50 to 80 percent: 2.0 points more demand served for 47 more units on the shelf. From 95 to 99: 0.0 points for 51 units. Above the 80 percent target, every step adds stock and no service in these 26 weeks.

→ Compare added stock with added service before raising the target.

# Same Target, Different Outcomes

Three products, the same 95 percent target, the same rule, replayed over the same 26 weeks.

| Product              | Lead time | Error spread (sd) | Fill rate | Stock-out weeks | Average stock, weeks of demand |
|----------------------|-----------|-------------------|-----------|-----------------|--------------------------------|
| Plush Bear, store    | 4 weeks   | 39.7 units        | 100.0 %   | 0               | 1.7                            |
| Puzzle 1000, store   | 5 weeks   | 27.5 units        | 87.2 %    | 6               | 1.9                            |
| Vacuum Filter, store | 6 weeks   | 0.92 units        | 93.3 %    | 1               | 7.5                            |

*Average stock on the shelf ÷ average weekly demand, 26 replay weeks.*

- **Plush Bear:** no stock-out, 1.7 weeks of demand on the shelf. Its forecast ran about 9 percent high, which added to the cushion
- **Puzzle:** a 95 percent target and only 87 percent of the units served. Its forecast ran about a third too low in the test months, and the safety stock covers only the spread of the misses, not a bias
- **Vacuum Filter:** 7.5 weeks of demand on the shelf, in about 4 units. Normal for a slow mover with a 6-week lead time; check the unit cost before calling it too much

→ **Inspect the product and its costs before calling stock excessive.**

# Hands-On 2: Choosing A Service Target

Open **Notebook 2**, sections 1 to 4. Copy to Drive, Run all, work in pairs.

- Section 1: the worked order, the same numbers as on the slides
- Section 2: move the target slider for the Plush Bear: 50, 80, 95, 99 percent. What does each step buy?
- Section 3: compare the three products at 95 percent. Then choose the Puzzle or the Filter in the section 2 slider: does the same target still make sense?
- Section 4: write your decision in four lines: the target you chose, the stock and fill rate you observed, what you accept, what you would check first
- Stuck? Runtime, Restart session and run all. Still stuck? Raise a hand, that is what I am here for

→ **Minimum: one product, two service targets, one defended choice.**

# Holding Cost Of Each Forecast

Plush Bear, store, 95 percent target, the same rule: only the forecast changes, and each method brings its own error spread.

| Forecast                     | Error spread (sd) | Average stock | Fill rate | Holding cost per year |
|------------------------------|-------------------|---------------|-----------|-----------------------|
| Repeat last year             | 77.7 units        | 351 units     | 100.0 %   | EUR 971               |
| Simple exponential smoothing | 59.6 units        | 208 units     | 88.0 %    | EUR 574               |
| LightGBM in global mode      | 39.7 units        | 289 units     | 100.0 %   | EUR 798               |

*Unit cost EUR 13.83, holding rate 20 percent of the unit cost per year. Shortage costs not included.*

- A better forecast has a smaller error spread, so it needs a smaller cushion: LightGBM in global mode serves all demand with 82 percent of the stock that Repeat last year needs
- Simple exponential smoothing holds the least stock because its flat forecast misses the Christmas rise: it runs out in all 4 December weeks and serves 88 percent of demand. It is cheaper to hold, but it is a different policy, not a better one

→ **Read stock, service and cost together.**

# Key Takeaways

- **Start with the pattern.** How often a product sells and how regular the amounts are tell you what error to expect and which methods to try, before any model
- **A benchmark in every table.** Keep repeating last year in every comparison: a method that cannot beat it is not worth its complexity. Over our test year, every method beat it
- **LightGBM in global mode, one model for all products, wins on every kind of demand** (smooth series: WAPE 19 % against 37 % for repeating last year), but runs low wherever sales are irregular. It learns from all products at once, including the planned promotions
- **Every accuracy number is blind somewhere.** MAPE is undefined when demand is zero and rewards low forecasts; WAPE hides small products. Never read accuracy without bias
- **Ask how it was tested.** How many test dates, which weeks of the year, and whether future information leaked in (leakage)
- **Features carry the knowledge.** A series becomes lags, averages, calendar and plan; tune the settings on validation weeks and keep the test year for the end
- **A model has reasons.** Ask for them as factors; "the model expects this December week to add about 22 percent" is a sentence you can check
- **Judge a forecast by the stock it needs.** Safety stock is  $z$  times the error spread over the protection period, so a better forecast means the same service with less stock
  
- **What will we order, and what evidence makes that a reasonable choice?**

# Further Reading And Tools

The notebooks and the data stay at [github.com/FH-Prevail/jku\\_course](https://github.com/FH-Prevail/jku_course). Most of the reading is free.

| If you want to                           | Start here                                                                                                                                                                                  |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open the notebooks again                 | <a href="#">Notebook 1 in Colab</a> and <a href="#">Notebook 2 in Colab</a> ; in Colab, File, Save a copy in Drive                                                                          |
| Understand the methods properly          | Hyndman, Athanasopoulos: Forecasting: Principles and Practice, 3rd ed., <a href="https://otexts.com/fpp3">otexts.com/fpp3</a> , free online                                                 |
| Read about slow movers                   | Syntetos, Boylan (2005), Int. J. Forecasting 21; Teunter, Syntetos, Babai (2011), EJOR 214; Mirshahi et al. (2026), Two-Step Framework for Intermittent Demand Forecasting, Procedia CS 278 |
| See a global model in a real competition | Makridakis, Spiliotis, Assimakopoulos (2022): The M5 competition, Int. J. Forecasting 38                                                                                                    |
| Try Prophet                              | <a href="https://facebook.github.io/prophet">facebook.github.io/prophet</a> ; Taylor, Letham (2018): Forecasting at Scale, The American Statistician 72                                     |
| Try a foundation model                   | <a href="https://github.com/NX-AI/tirex">github.com/NX-AI/tirex</a> ; my comparison: Mirshahi et al. (2026), Forecasting Without Fine-Tuning, Procedia CS 277                               |
| Read about promotions in retail          | Darbanian, Brandtner, Falatouri, Nasseri, Mirshahi (2025): Timing Matters, J. Retailing and Consumer Services 85                                                                            |
| Go deeper on inventory                   | Silver, Pyke, Thomas: Inventory and Production Management in Supply Chains, chapters on safety stock and on the newsvendor problem (one order for one selling period)                       |
| Ask me anything                          | <a href="mailto:Sina.Mirshahi@fh-steyr.at">Sina.Mirshahi@fh-steyr.at</a>                                                                                                                    |

# Thank you



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Notebooks and data:  
[github.com/FH-Prevail/jku\\_course](https://github.com/FH-Prevail/jku_course) →



# Reference: Exponential Smoothing

Weighted averages that trust the newest weeks most; Holt-Winters adds a trend and a season.

| Method                                    | What it tracks              | Update rule                                                            | Forecast shape                          | WAPE, smooth series |
|-------------------------------------------|-----------------------------|------------------------------------------------------------------------|-----------------------------------------|---------------------|
| <b>SES (simple exponential smoothing)</b> | the level                   | new level = $\alpha$ x this week's sales + (1 - $\alpha$ ) x old level | flat                                    | 29.4 %              |
| <b>Holt</b>                               | level and trend             | the same rule again for the week-to-week change                        | a straight line; damped: flattens out   | not in our backtest |
| <b>Holt-Winters</b>                       | level, trend, yearly season | a third rule for each week of the year; needs two full years           | last year's shape around a moving level | 32.4 %              |

- $\alpha$  (alpha) is the weight of the newest week, between 0 and 1, fitted from each product's history: 0.24 for the Plush Bear, but close to 0 for 65 of our 80 series, whose forecast is then almost the plain average of all past weeks
- **WAPE**, the total miss as a percentage of total demand (lower is better; chapter 3), on the 26 smooth series over 2025: Holt-Winters 32.4 %, SES 29.4 %, seasonal naive 36.6 %. A season learned from one product's two years is a noisy season

→ Adding components does not automatically help: here the flat SES beats Holt-Winters four weeks ahead.



# Reference: Slow Movers

Smoothing a series that is zero most weeks gives a forecast that jumps after every sale. Croston fixed that in 1972.

| Method                                   | Idea                                                                                    | When a product stops selling | WAPE, intermittent | Bias |
|------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------|--------------------|------|
| <b>Croston</b>                           | Estimate the size of a sale and the gap between sales separately. Forecast = size ÷ gap | Keeps the old rate forever   | 129.9 %            | -4 % |
| <b>Croston SBA (Syntetos and Boylan)</b> | Croston with a small downward correction, because in theory Croston runs slightly high  | Same as Croston              | 128.4 %            | -8 % |
| <b>TSB (Teunter, Syntetos, Babai)</b>    | Update the chance of a sale every week, the size only when a sale happens               | Lowers the rate week by week | 132.3 %            | -4 % |

- **Bias** (forecast minus actual, as a share of actual; minus means too low): Croston -4 %, TSB -4 %, SBA -8 %: on our data the correction goes the wrong way. LightGBM in global mode: -15 %
- **WAPE** (the total miss as a percentage of total demand, lower is better; chapter 3) on the same 23 intermittent series: SES 129 %, Croston, SBA and TSB 128 to 132 %, LightGBM in global mode 121 %. Above 100 percent is normal for a product that sells one unit some weeks
- **Our own research:** UR2CUTE (2024) forecasts a slow mover in two steps with two neural networks: first whether a week has a sale, then how many units (Mirshahi, Brandtner, Komínková Oplatková 2024, MENDEL)

→ **The forecast is a weekly rate: add it up over the weeks until an order arrives (the lead time), and judge it by the stock decision.**

# Reference: One Christmas Order

Newsvendor: one order for one selling period. Puzzle 1000, store, the week of 15 December 2025.

| Cost recovered on a leftover | Cost of a leftover | Critical ratio | Order     |
|------------------------------|--------------------|----------------|-----------|
| 0 % of the purchase cost     | EUR 38.86          | 0.37           | 80 units  |
| 50 % of the purchase cost    | EUR 19.43          | 0.54           | 92 units  |
| 95 % of the purchase cost    | EUR 1.94           | 0.92           | 128 units |

- Forecast **88 units**, error spread 27. A unit short costs the lost margin, EUR 23.23
  - Critical ratio = cost short ÷ (cost short + cost of a leftover). The order is the demand level that is not exceeded with that probability: a **quantile**
  - Actual demand that week: 41 units
- If you do not know the costs, you cannot recommend a quantity

→ The forecast stays fixed; the cost of leftovers changes the order.

# Reference: The Limits Of Our Replay

- The sales are artificial; the four flagged stock-out weeks were estimated for training and left out of the scoring
- Orders are whole units; supplier lead times are fixed; deliveries arrive as scheduled; there are no backorders: demand we cannot serve is lost, not delivered later
- The safety stock rule assumes stable, unbiased, independent misses that follow a bell curve. For sparse demand that is a poor assumption
- The error spread was measured on 25 weeks and the rule replayed on 26 weeks; costs, bias and changing conditions still need review
- LightGBM in global mode needs a year of history per product; it cannot forecast a new product that has no sales history yet

→ **A reproducible classroom result is evidence, not a guarantee for a retailer.**